CIRCUITOS DIGITALES Y MICROCONTROLADORES 2025

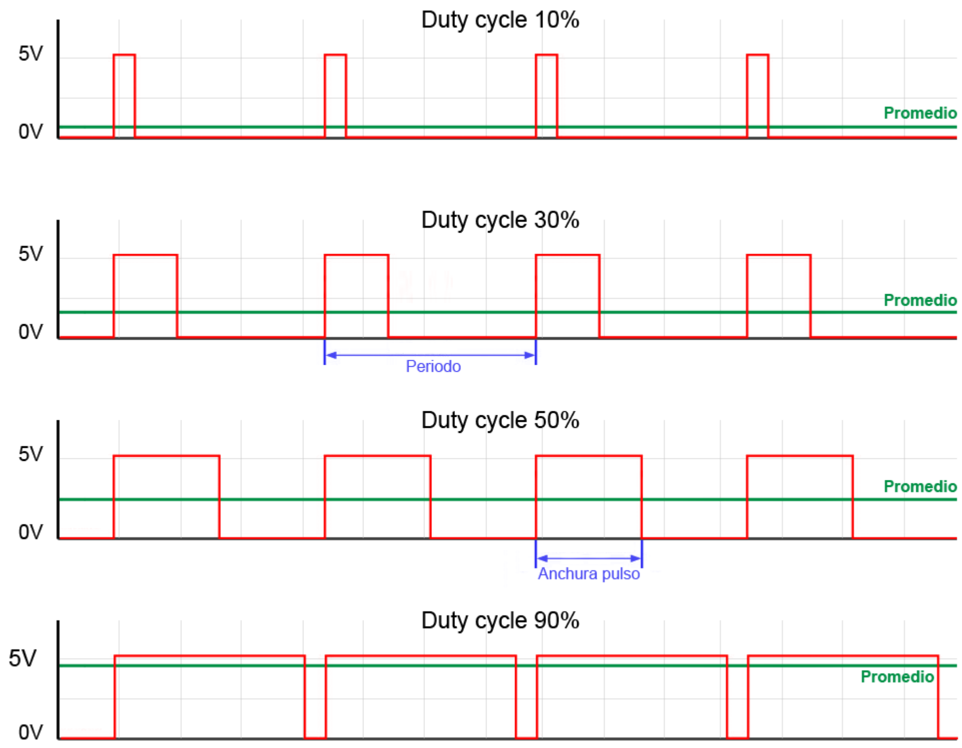
Facultad de Ingeniería
UNLP

PERIFERICOS TIMER
Modos PWM

Ing. José Juárez

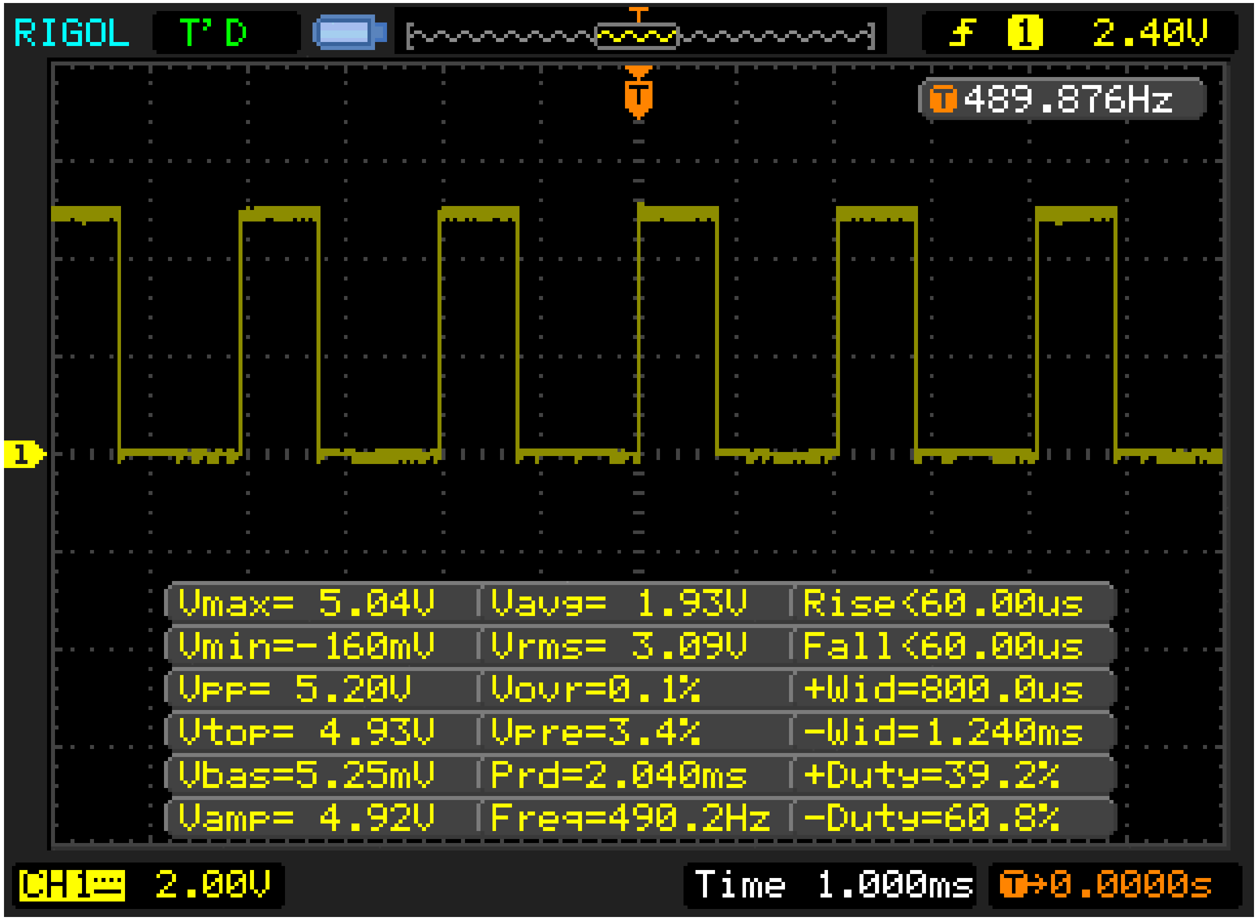
Señales PWM (Pulse Width Modulation)

- Es una técnica de modulación digital donde la información útil de la señal se encuentra en el ancho del pulso. Esto permite que se pueda obtener una señal analógica a partir de una señal digital.



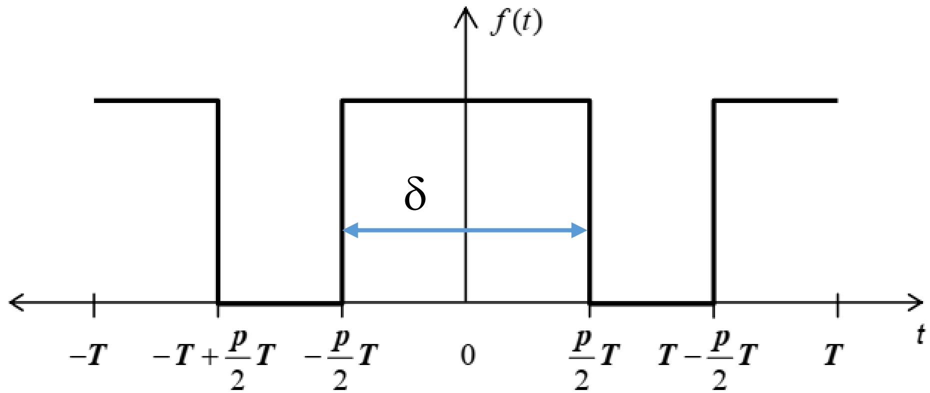
- Permite entre otras aplicaciones, controlar dispositivos analógicos por medio de salidas digitales.
- Ej.: control de velocidad de motores DC, control de intensidad de lámparas, Control de color en LED RGB.

PWM ejemplos



PWM análisis de Fourier

- Sea f(t) una señal PWM de periodo T y ciclo de trabajo $0 \leq p \leq 1$



p= δ/T

- La representación por Series de Fourier de la señal es:

$$f(t) = A_0 + \sum_{n=1}^{\infty}\left[A_n \cos\left(\frac{2n\pi t}{T}\right) + B_n \sin\left(\frac{2n\pi t}{T}\right)\right]$$

$$A_n = \frac{1}{T}\int_{-T}^{T} f(t)\cos\left(\frac{2n\pi t}{T}\right)dt$$

$$A_0 = \frac{1}{2T}\int_{-T}^{T} f(t)dt$$

$$B_n = \frac{1}{T}\int_{-T}^{T} f(t)\sin\left(\frac{2n\pi t}{T}\right)dt$$

Valor medio o componente DC

armónicos

PWM análisis de Fourier

- Si la amplitud de la señal es $V_{OH}=V_{DD}$

$$A_0 = V_{DD}\, p$$

Información de interés!
El valor medio de la señal es proporcional al ciclo de trabajo p

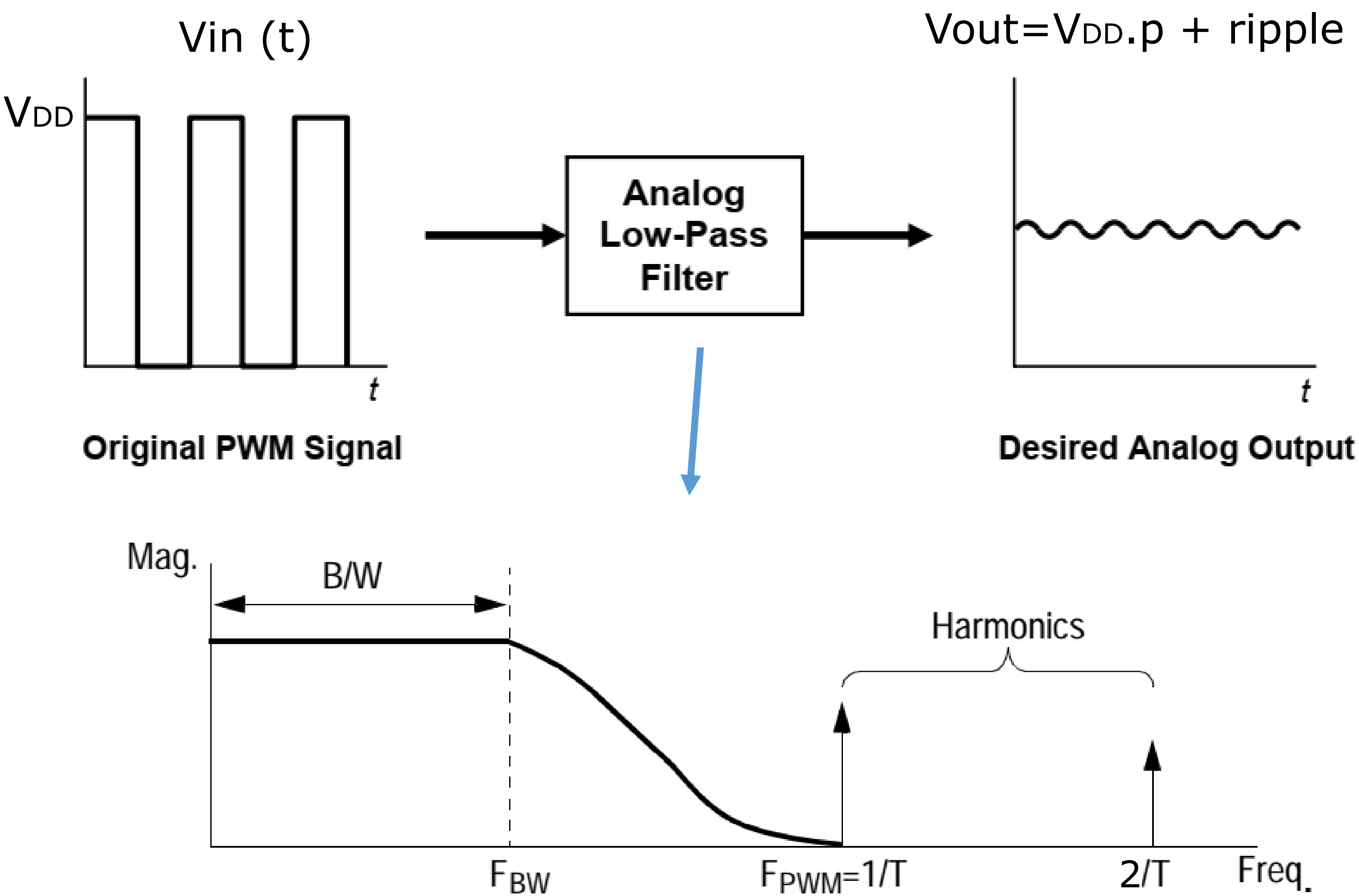
$$A_n = V_{DD} \frac{1}{n\pi}\left[\sin(n\pi p) - \sin(2n\pi(1-p/2))\right]$$

Amplitud de la n-ésima armónica

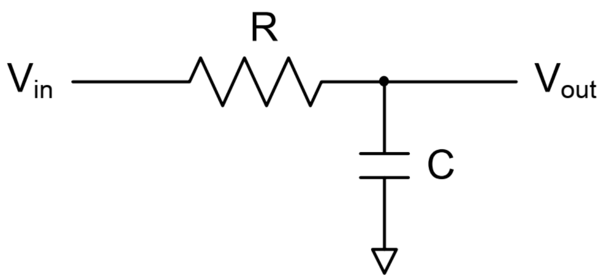
$$B_n = 0$$

- La señal PWM contendrá una componente de DC de amplitud A0 (valor medio de la señal) y las armónicas de la frecuencia PWM a $f=n*f_{PWM}$ (con n entero) y de amplitud An respectivamente.

PWM Filtrado



PWM Filtrado

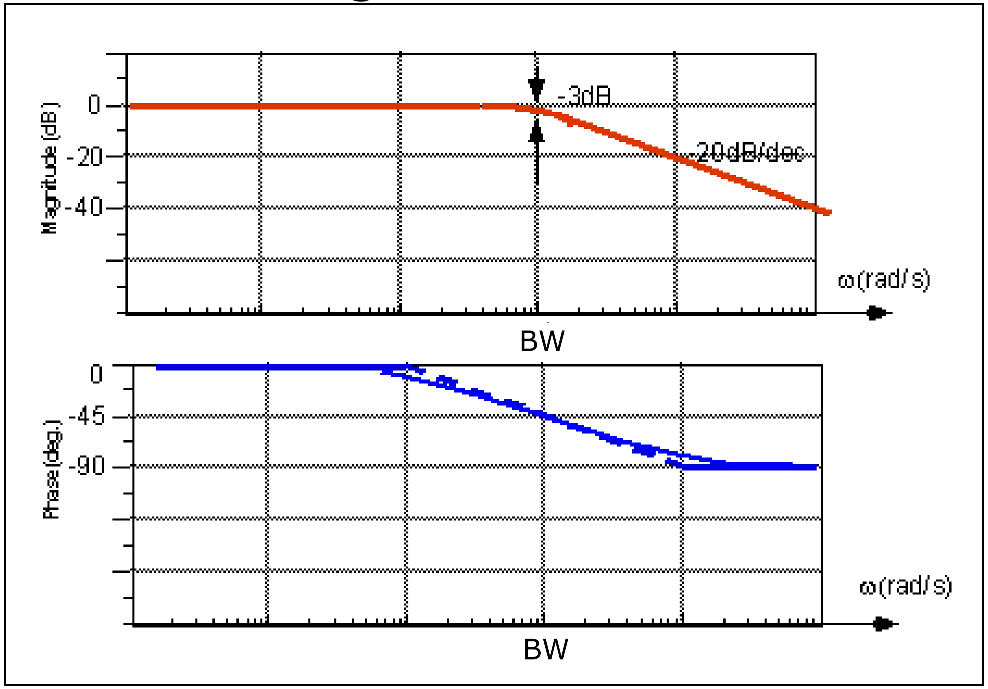


$$H(s)=\frac{V_{out}(s)}{V_{in}(s)}=\frac{1}{\tau s+1}$$

τ=RC $\qquad$ s=jω

$$BW=\frac{1}{\tau} \quad \text{(rad/s)}$$

Diagrama de Bode



$$dB = 10\log\left(\frac{V_{out}}{V_{in}}\right)^2 = 10\log|H(f)|^2$$

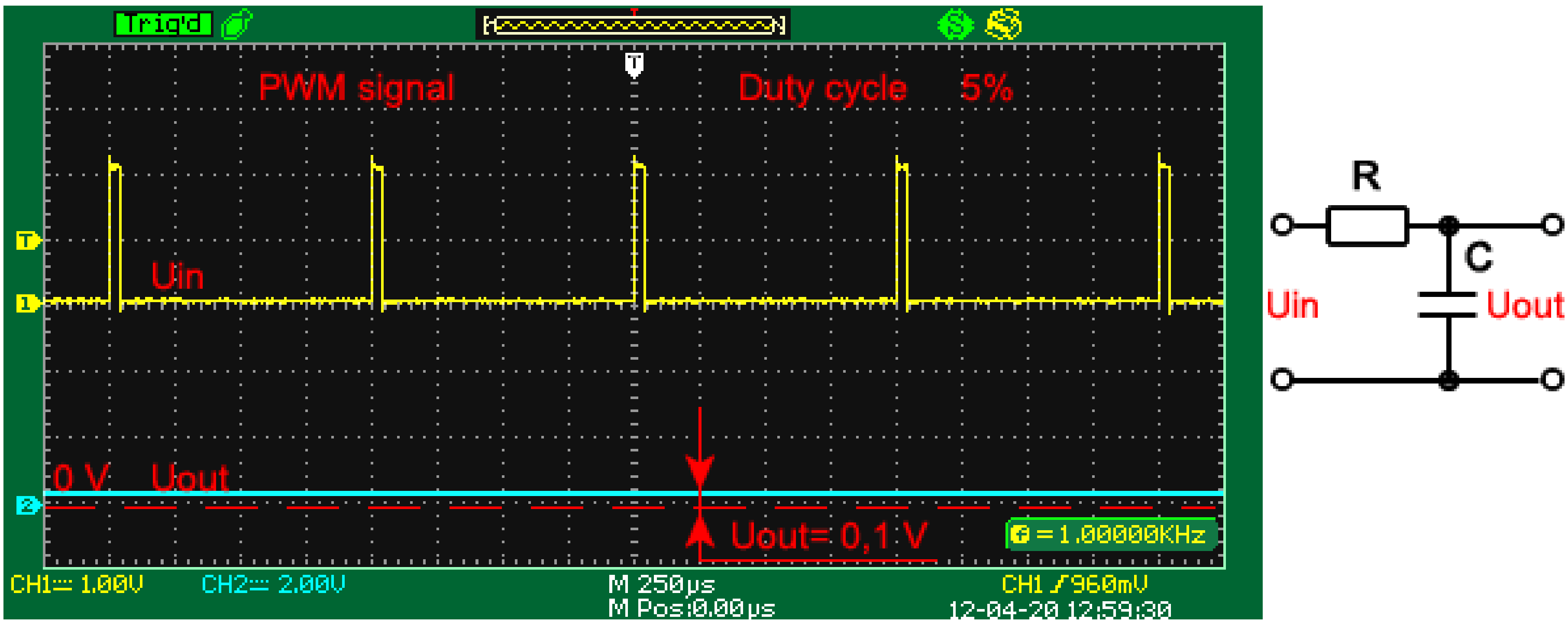
$$|H(f)| = \frac{1}{\sqrt{1+(2\pi \times f \times RC)^2}}$$

$\phi(H(f))$=-arctg($2\pi f RC$)

$V_{out}(t) \cong A_0 + A_1|H(f_{pwm})|\cos(2\pi f_{pwm} t + \phi(f_{pwm}))$

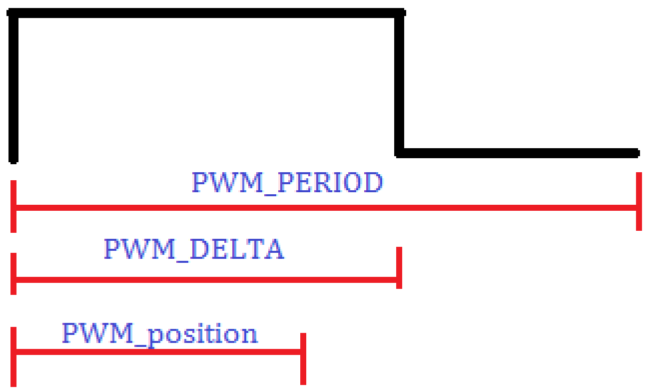
Si hacemos BW= ωpwm/10 entonces la primera armónica será atenuada 20dB es decir 10 veces!

PWM Filtrado



http://sim.okawa-denshi.jp/en/CRlowkeisan.htm

Generación de PWM por Software



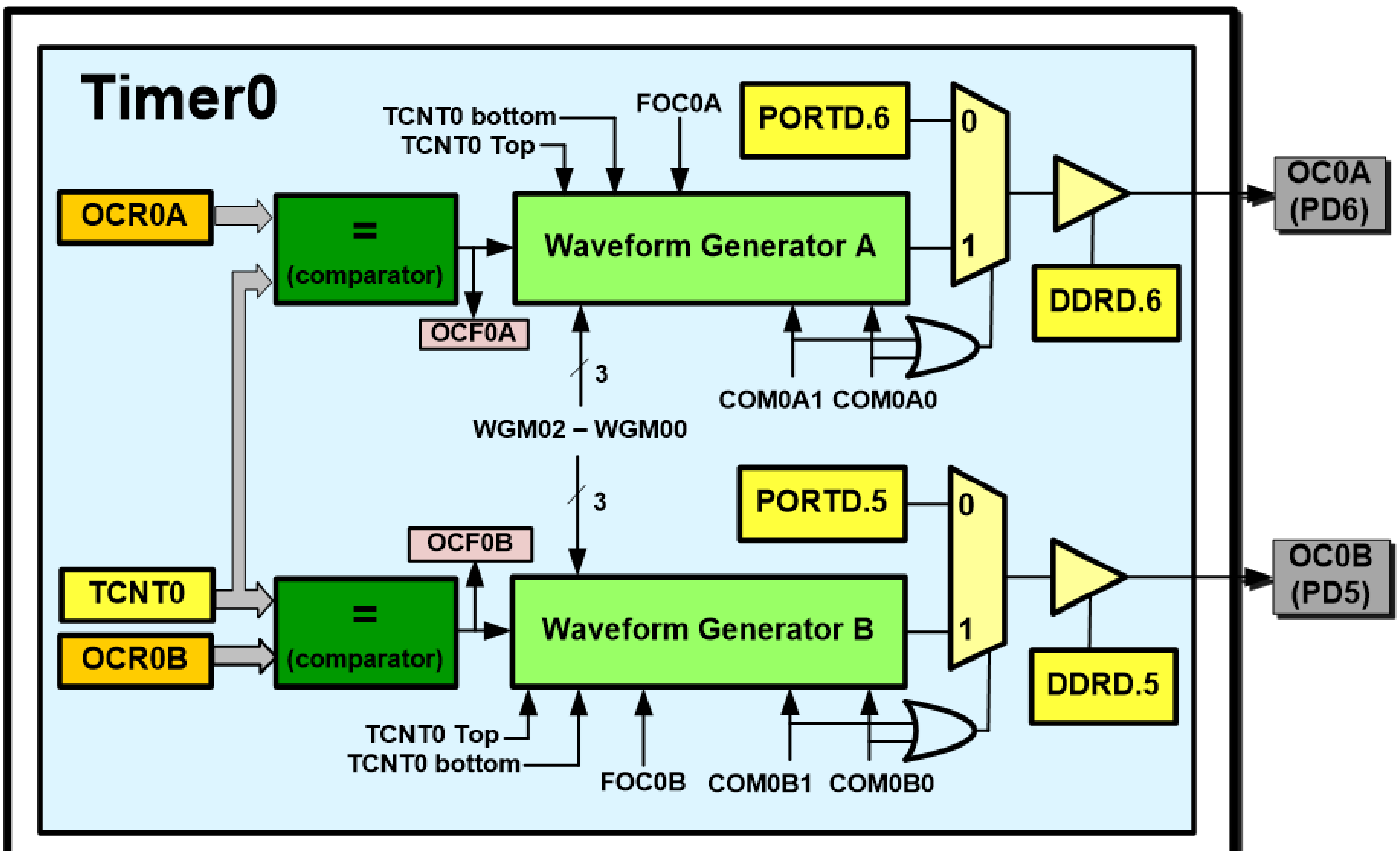

```c
#define PWM_PERIOD  100
#define PWM_DELTA   25
#define PWM_OFF     PORTB &=~(1<<PORTB0)
#define PWM_ON      PORTB |=(1<<PORTB0)
#define PWM_START   DDRB |=(1<<PORTB0)

// interrupción períodica cada 1ms
ISR (TIMER0_COMPA_vect)
{
  PWM_soft_Update();
}
```

```c
void PWM_soft_Update(void){
static uint16_t PWM_position=0;
    if (++PWM_position>=PWM_PERIOD){
        PWM_position=0;
    }
    if (PWM_position<PWM_DELTA){
    PWM_ON;
    } else{
    PWM_OFF;
    }
}
```

¿cual es la frecuencia de la señal PWM generada en PB0?
¿cual es el ciclo de trabajo?

Generación de PWM por Hardware



PWM – Timer 0

- Recordemos los registros de configuración TCCR0A y B

	7	6	5	4	3	2	1	0
TCCR0A	COM0A1	COM0A0	COM0B1	COM0B0	–	–	WGM01	WGM00
	R/W	R/W	R/W	R/W	R	R	R/W	R/W

Compotamiento de los generadores de señal A y B

Modos

	7	6	5	4	3	2	1	0
TCCR0B	FOC0A	FOC0B	–	–	WGM02	CS02	CS01	CS00
	W	W	R	R	R/W	R/W	R/W	R/W

Fuente y predivisor de reloj

PWM – Timer 0

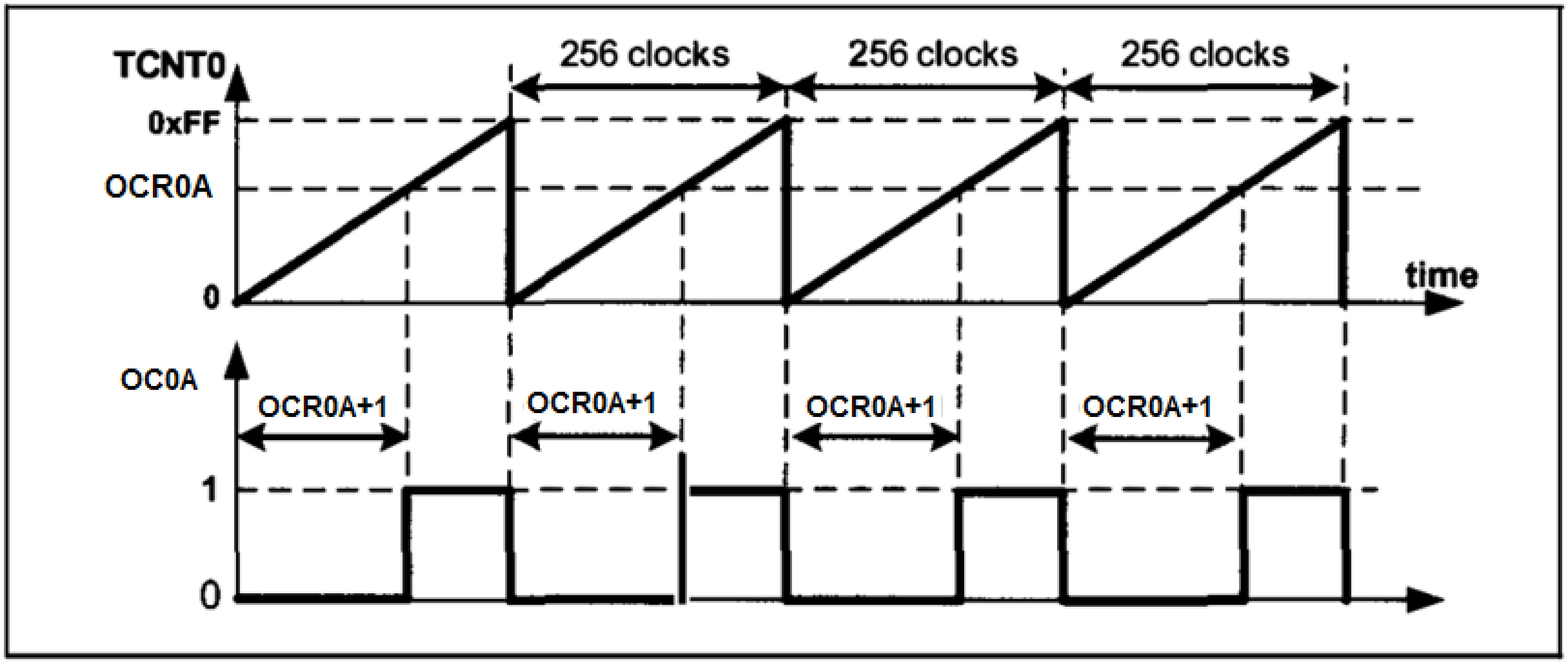
Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on[(1)(2)]
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

Notes: 1. MAX = 0xFF
2. BOTTOM = 0x00

Generador PWM – Timer 0

- Modo Fast PWM modo 3: operación



Modo Fast PWM Modo Invertido

PWM – Timer 0

Table 14-3. Compare Output Mode, Fast PWM Mode[1]

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	WGM02 = 0: Normal Port Operation, OC0A Disconnected. WGM02 = 1: Toggle OC0A on Compare Match.
1	0	Clear OC0A on Compare Match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on Compare Match, clear OC0A at BOTTOM, (inverting mode).

Modo Fast PWM: invertido, no invertido

PWM – Timer 0

- Utilizaremos el generador de onda y la salida de comparación con las siguientes posibilidades:

COM0A1=1
COM0A0=0
no invertido

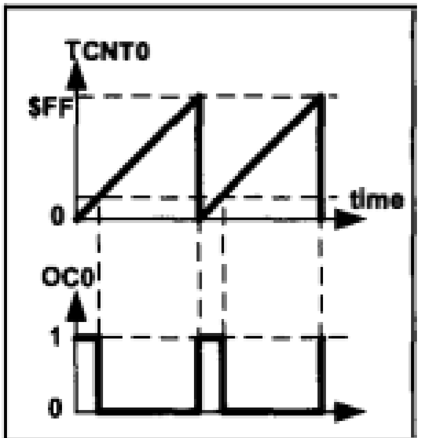


Figure 16-13A. Non-inverted

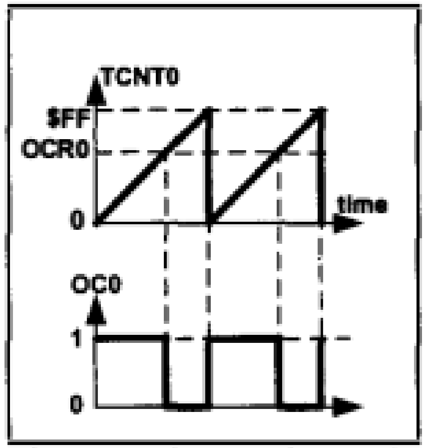


Figure 16-13B. Non-inverted

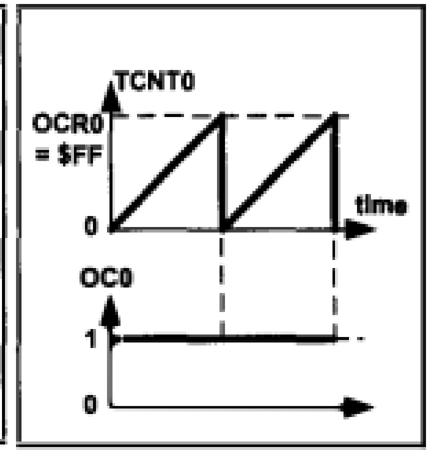


Figure 16-13C. Non-inverted

COM0A1=1
COM0A0=1
invertido

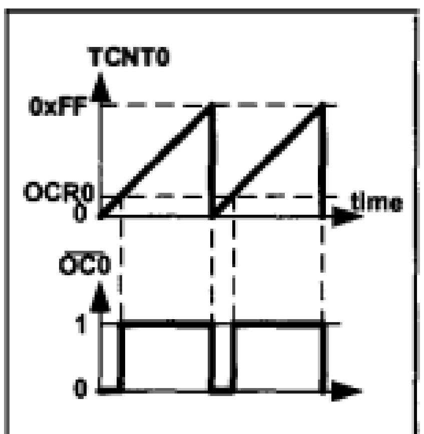


Figure 16-14A. Inverted

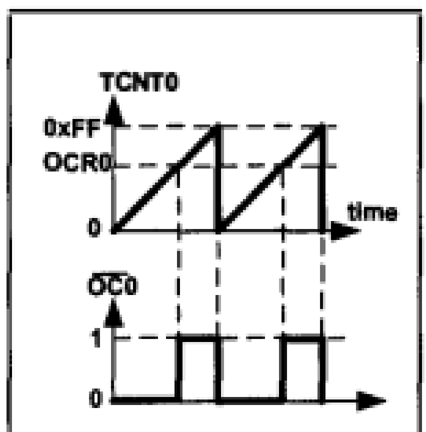


Figure 16-14B. Inverted

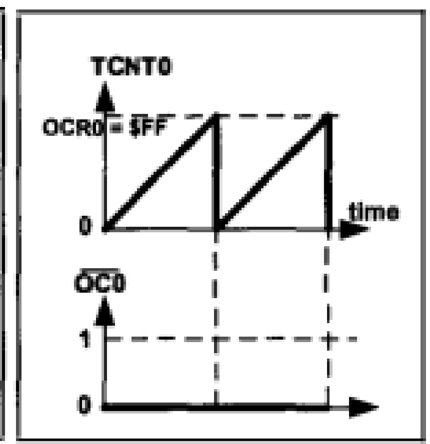
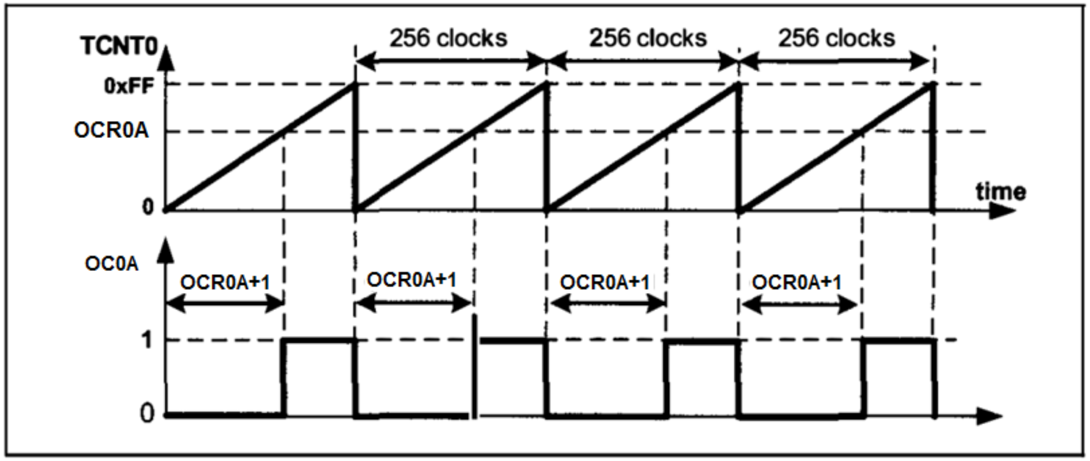


Figure 16-14C. Inverted

PWM – Timer 0



- Frecuencia de la señal generada Modo 3:

$$\left.\begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{256} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned}\right\} \Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{256 \times N}$$

- Ciclo de trabajo: $p = \delta/T$

$$\text{Duty Cycle} = \frac{OCR0 + 1}{256} \times 100$$

Modo no invertido

$$\text{Duty Cycle} = \frac{255 - OCR0}{256} \times 100$$

Modo invertido

PWM – Timer 0

- Ejemplo: Generar pwm con f=7812.5Hz y CT=37.5%

  - Supongamos fclk=16MHz

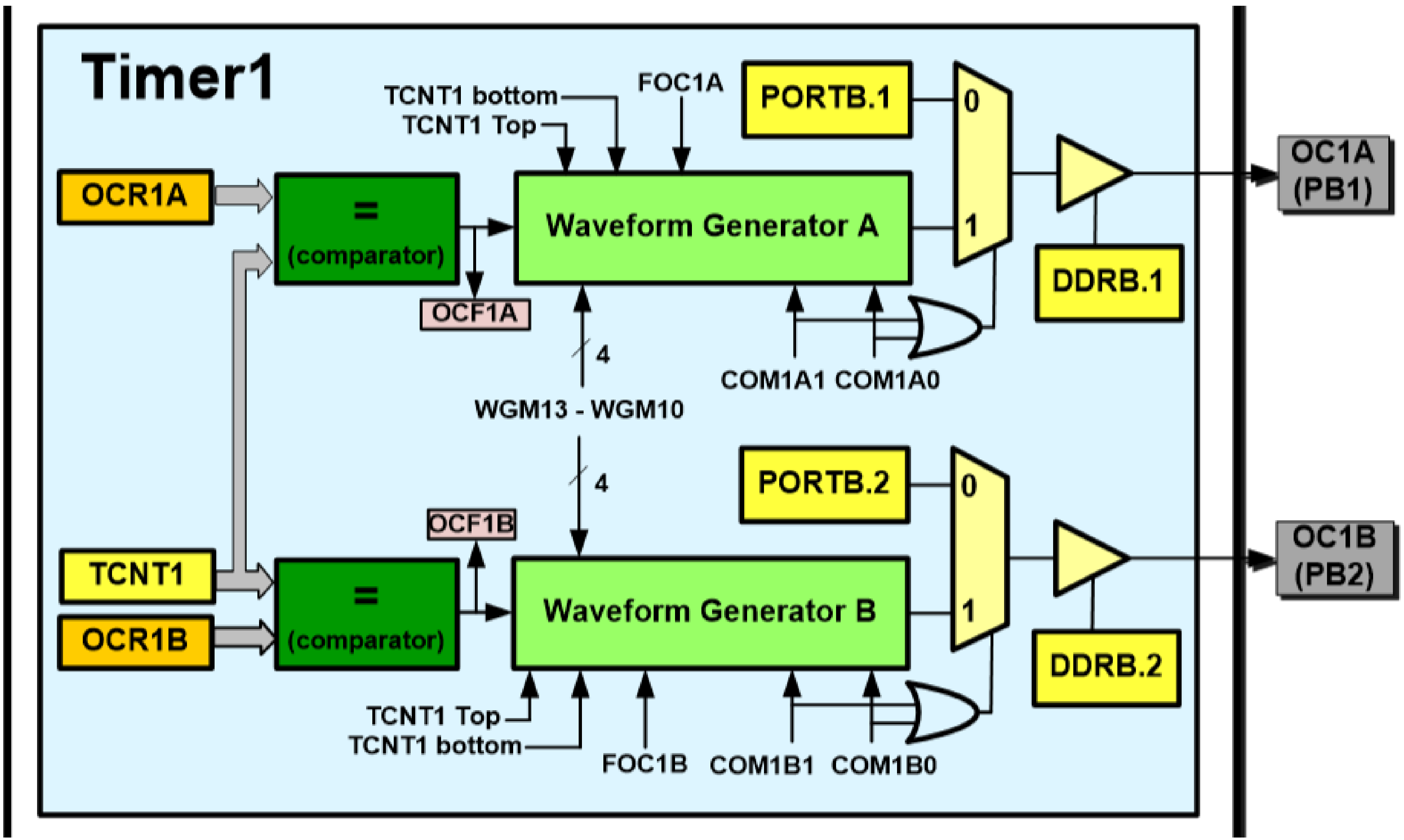
$\Rightarrow$ duty cycle=(OCR0A+1)*100/256 = 37.5% despejando OCR0A=95

$\Rightarrow$ si N=8 fpwm=f_{clk}/(256*N) = 7812.5Hz

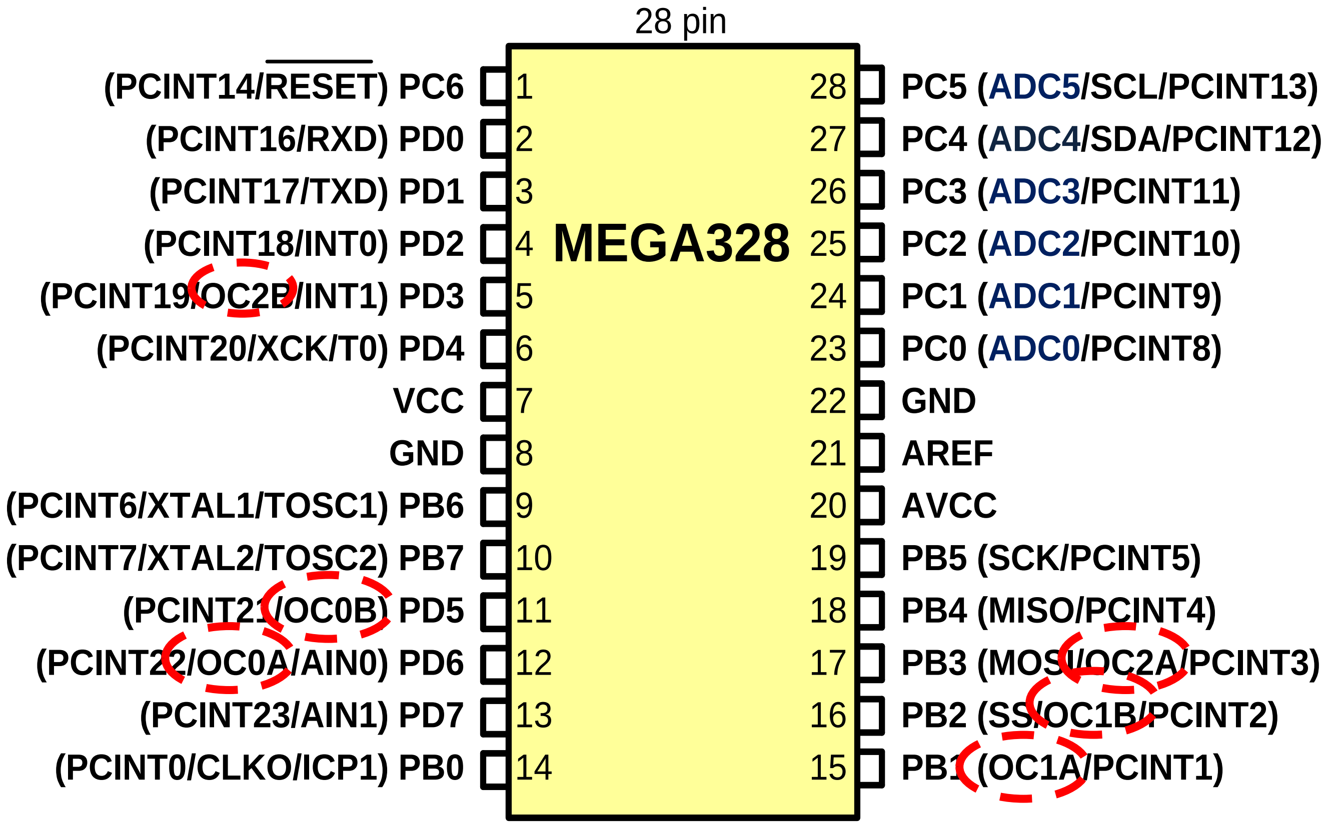
$\Rightarrow$Código:

```
DDRD |=(1<<6);  //PIN PD6 corresponde a la salida OC0A
OCR0A=95;
TCCR0A=(1<<COM0A1)|(1<<WGM01)|(1<<WGM00);    //modo Fast-non inv.
TCCR0B=(1<<CS01);                            //prediv 8
```

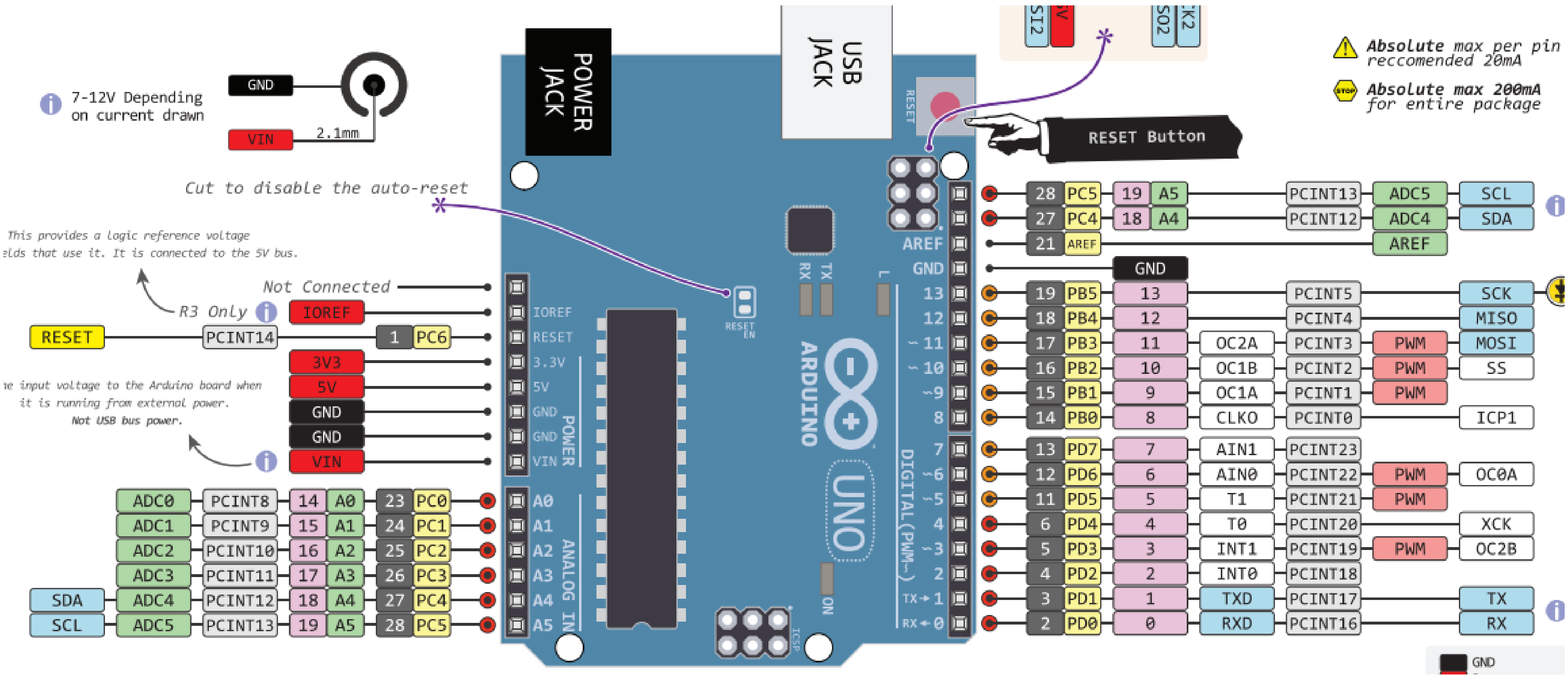
PWM – Timer 1 (16 bits)



PWM – Atmega328p



PWM – ARDUINO board



PWM – Timer 1 (16 bits)

- Modos Fast PWM

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	–	–	–
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

PWM – Timer 1 (16 bits)

- 5 Modos Fast PWM dependiendo del valor TOP del contador:

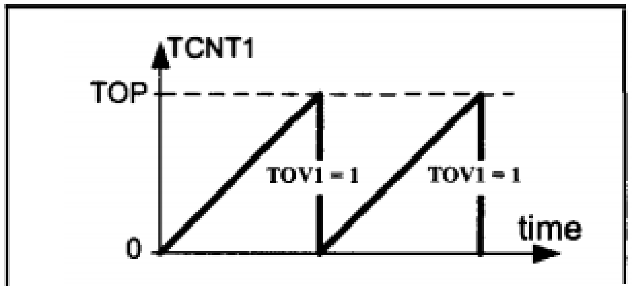


Figure 16-24. Fast PWM Mode

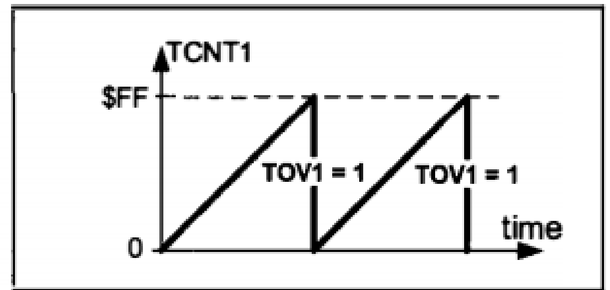


Figure 16-25. TOV in Mode 5

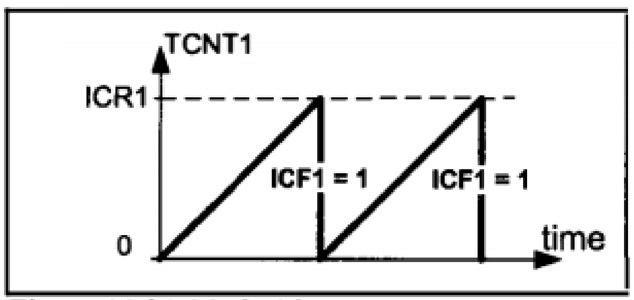


Figure 16-26. Mode 14

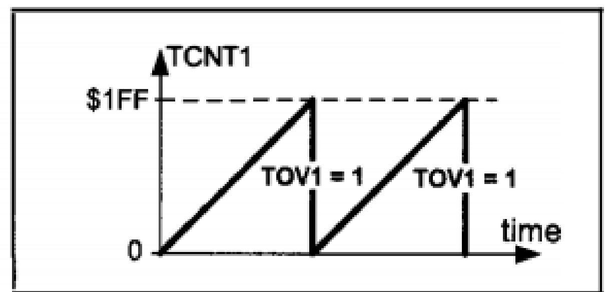


Figure 16-27. TOV in Mode 6

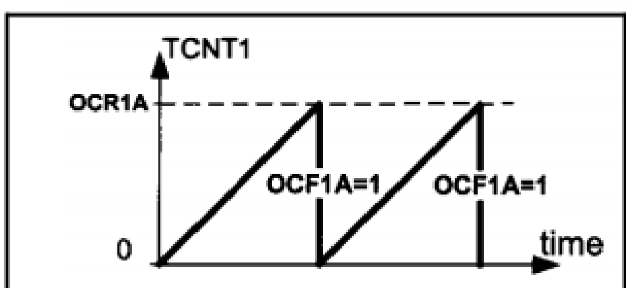


Figure 16-28. Mode 15

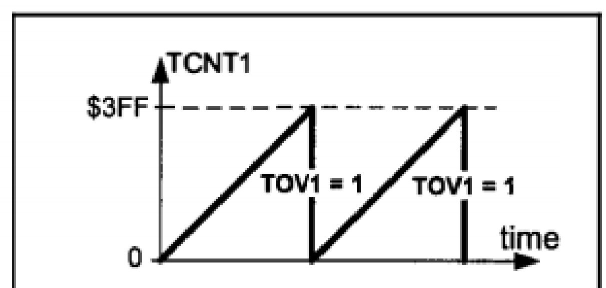


Figure 16-29. TOV in Mode 7

PWM – Timer 1 (16 bits)

- Modo no invertido

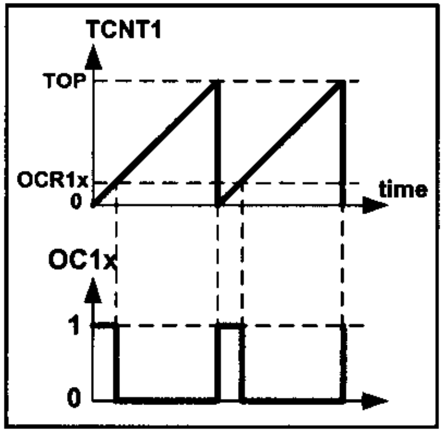


Figure 16-32A. Non-inverted

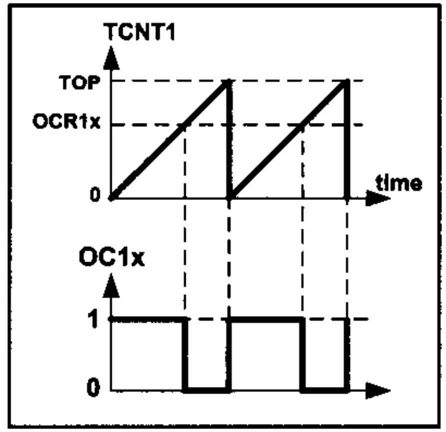


Figure 16-32B. Non-inverted

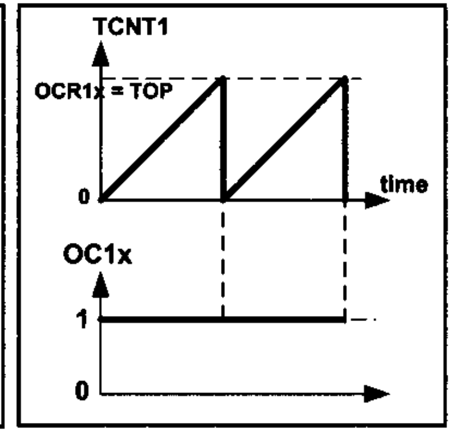


Figure 16-32C. Non-inverted

- Ídem Modo invertido
- Frecuencia fast pwm:

Número de bits de un PWM

$$R_{FPWM} = \frac{\log(TOP+1)}{\log(2)}$$

$$\left.\begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{\text{Top}+1} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned}\right\} \Longrightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{(\text{Top}+1)\times N}$$

PWM – Timer 1 (16 bits)

- Ciclo de trabajo: $p = \delta/T$

No invertido:

$$\boxed{\text{Duty Cycle} = \frac{\text{OCR1x} + 1}{\text{Top} + 1} \times 100}$$

Invertido:

$$\boxed{\text{Duty Cycle} = \frac{\text{Top} - \text{OCR1x}}{\text{Top} + 1} \times 100}$$

Ejemplo: para ciclo de trabajo 75%, el OCR1A debe ser:

- 191 (en modo 5,no inv)
- 383 (en modo 6,no inv)
- 767 (en modo 7,no inv)

PWM – Timer 1 (16 bits)

- Ejemplo: Generar PWM no inv. M5, f=31.250Hz y cicloT=75%.
  - Sup fclk=8MHz

```
DDRB |=(1<<PB1);      //PIN PB1 corresponde a la salida OC1A
OCR1A=191;
TCCR1A=(1<<COM1A1)|(1<<WGM10);  //modo 5 Fast-non inv.
TCCR1B=(1<<CS10)|(1<<WGM12);    //prediv 1
```

- Notar que las posibles frecuencias a generar dependen del prescalador

Prescaler	1	1:8	1:64	1:256	1:1024
Mode = 5	$\frac{F_{oscillator}}{1\times256}$	$\frac{F_{oscillator}}{8\times256}$	$\frac{F_{oscillator}}{64\times256}$	$\frac{F_{oscillator}}{256\times256}$	$\frac{F_{oscillator}}{1024\times256}$
Mode = 6	$\frac{F_{oscillator}}{1\times512}$	$\frac{F_{oscillator}}{8\times512}$	$\frac{F_{oscillator}}{64\times512}$	$\frac{F_{oscillator}}{256\times512}$	$\frac{F_{oscillator}}{1024\times512}$
Mode = 7	$\frac{F_{oscillator}}{1\times1024}$	$\frac{F_{oscillator}}{8\times1024}$	$\frac{F_{oscillator}}{64\times1024}$	$\frac{F_{oscillator}}{256\times1024}$	$\frac{F_{oscillator}}{1024\times1024}$

PWM – Timer 1 (16 bits)

- Para generar otras frecuencias se utilizan los modos 14 o 15 donde TOP se especifica con los registros ICR1 o OCR1A respectivamente.
- Por ejemplo: generar fast pwm, modo 14, f=80kHz

80kHz=8MHz / N*(TOP+1) => / N*(TOP+1) = 8MHz/80kHz = 100

Si N=1 (CS1[2:0]=001)

TOP+1=100

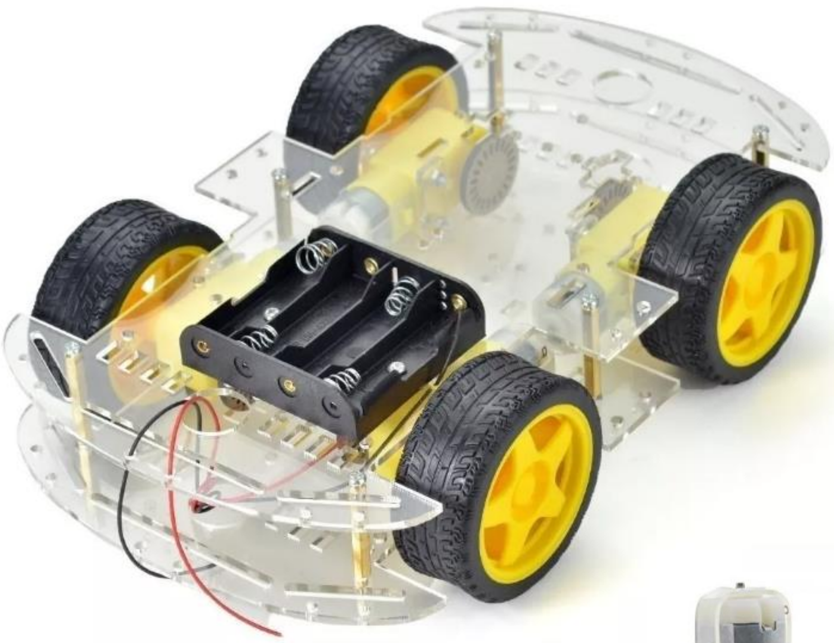
TOP=99

- Para ciclo de trabajo 20%

20=(OCR1x+1)*100 /(99+1) => OCR1x+1=20 => OCR1x =19

```
DDRB |=(1<<PB1);     //PIN PB1 corresponde a la salida OC1A
ICR1A=99;
OCR1A=19;
TCCR1A=(1<<COM1A1)|(1<<WGM11);  //modo 14 Fast-non inv.
TCCR1B=(1<<CS10)|(1<<WGM13)|(1<<WGM12);    //prediv 1
```

Aplicaciones PWM – control de Motor DC

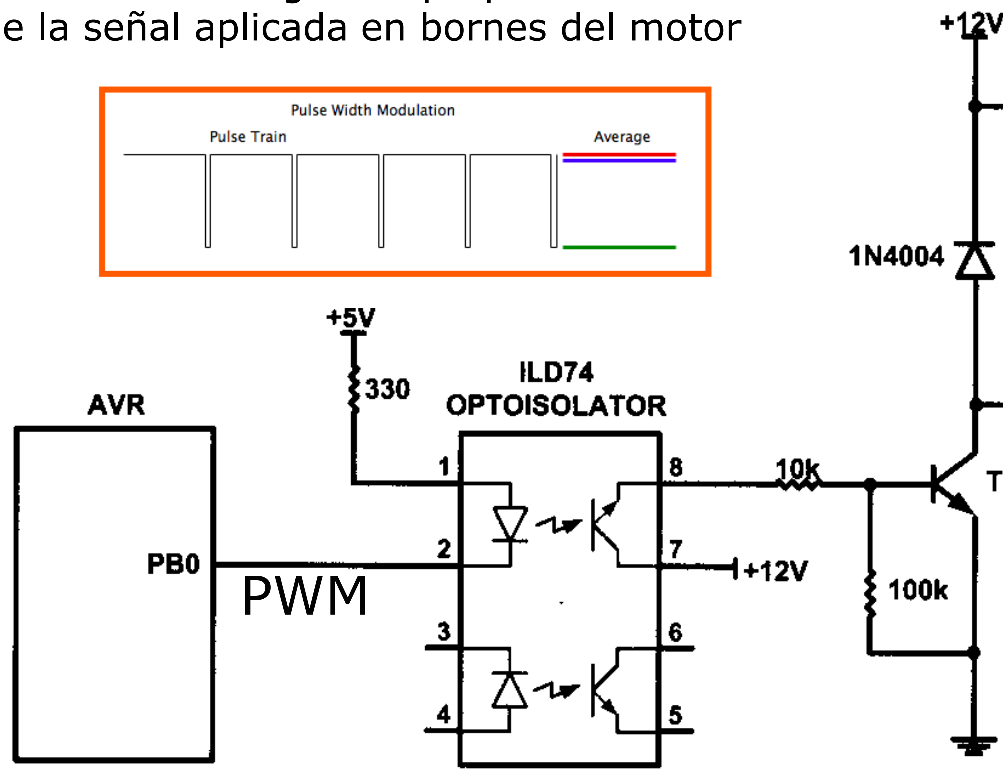


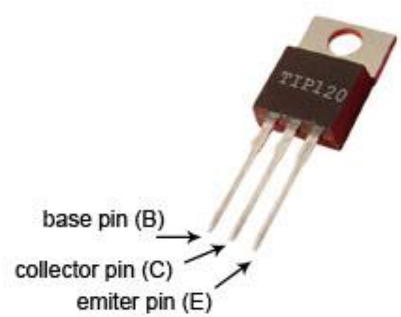
https://controlautomaticoeducacion.com/arduino/pwm-arduino/

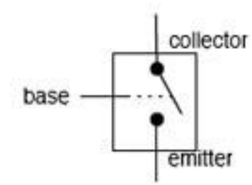
PWM – control de Motor DC

- Ejemplo de aplicación: control de velocidad unidireccional

La velocidad de giro es proporcional al valor medio
de la señal aplicada en bornes del motor

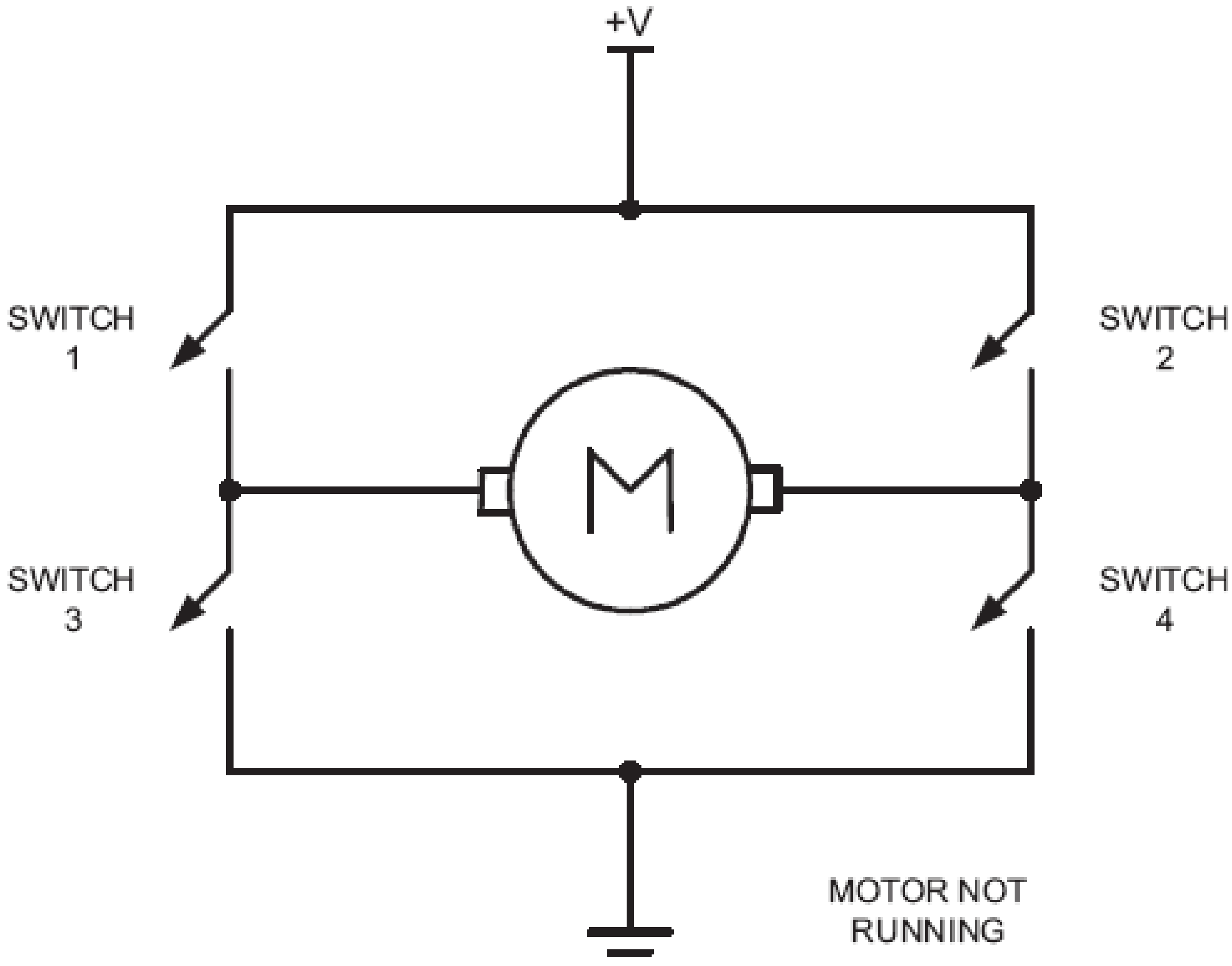




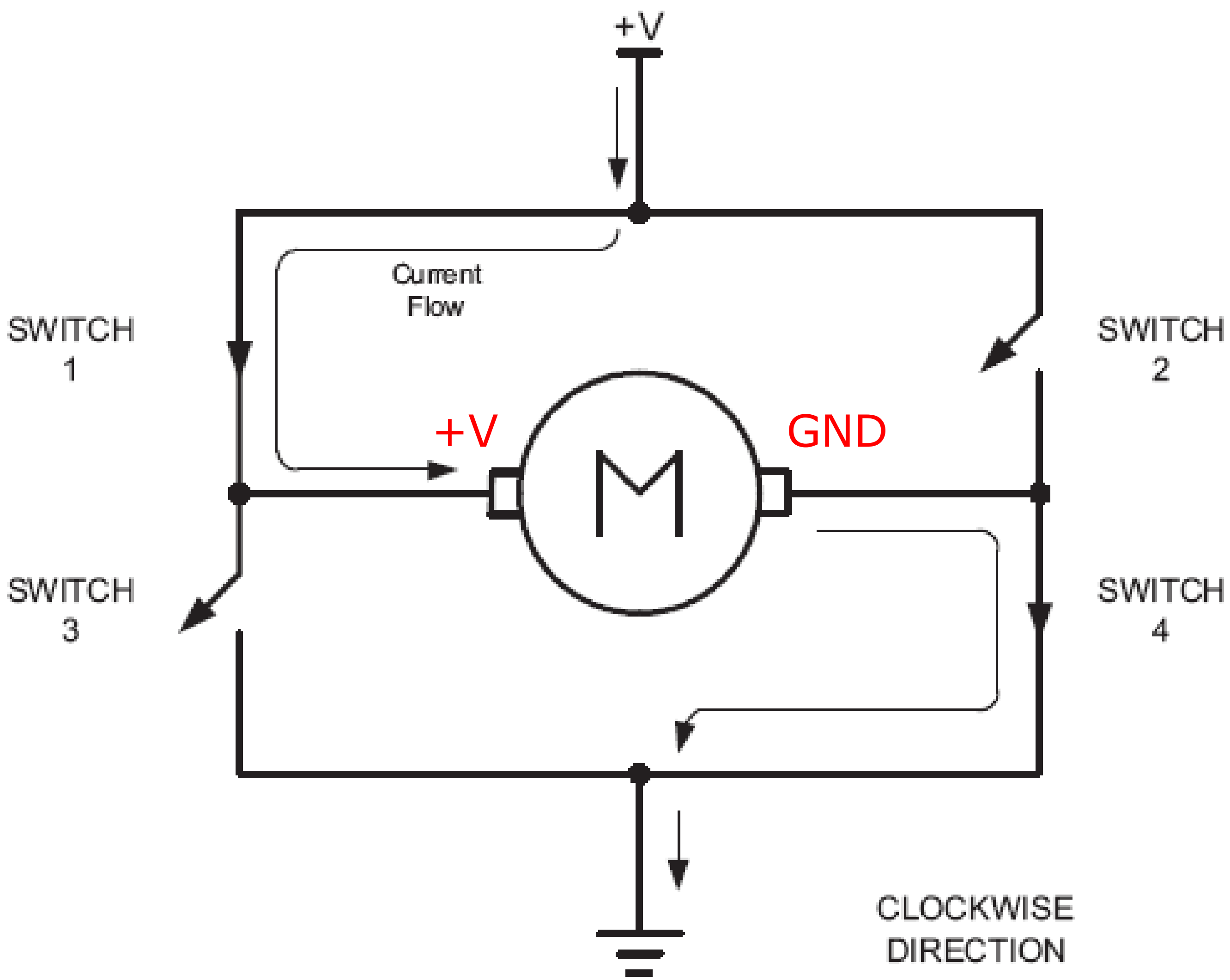


It works like a switch

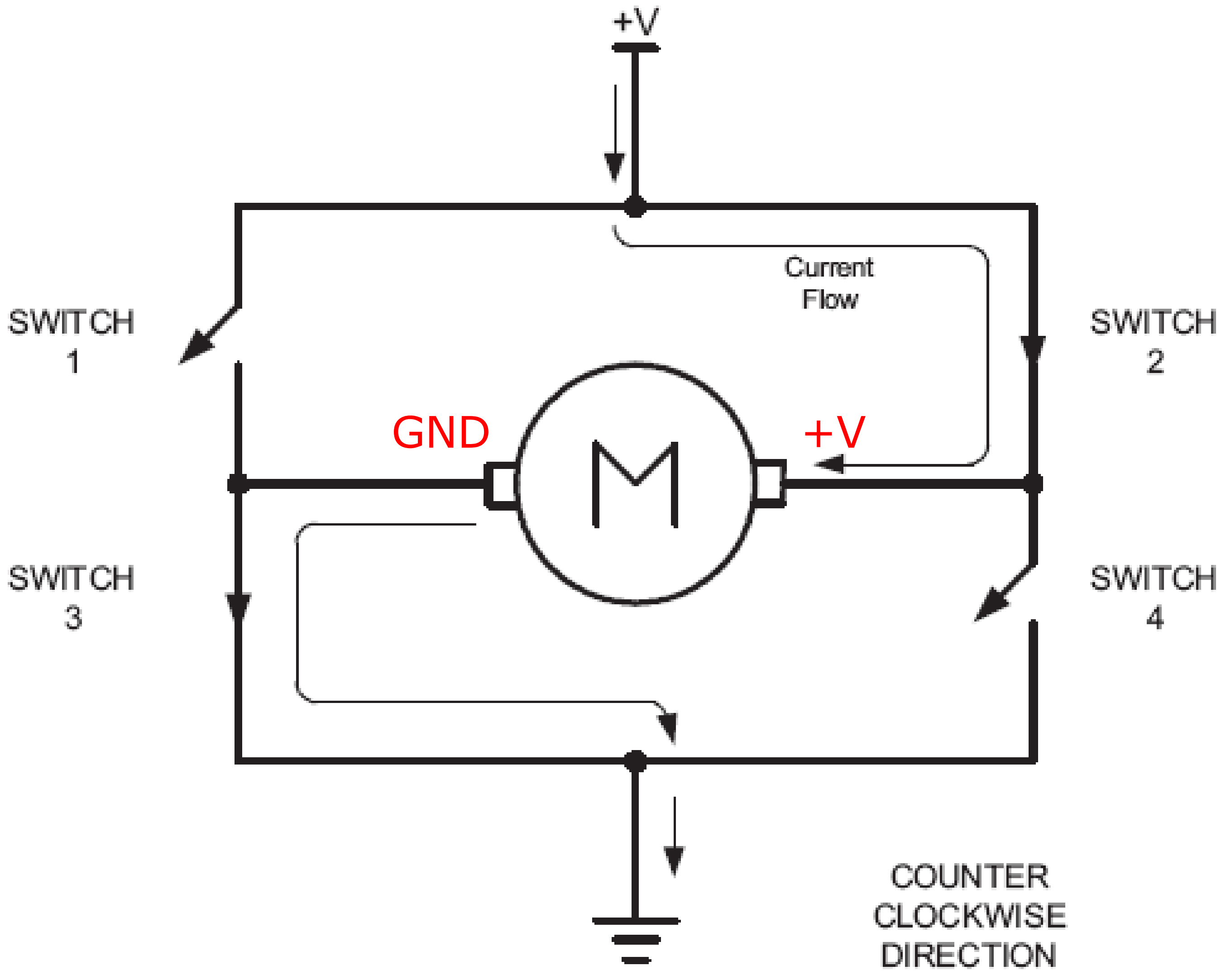
Motor DC control bidireccional



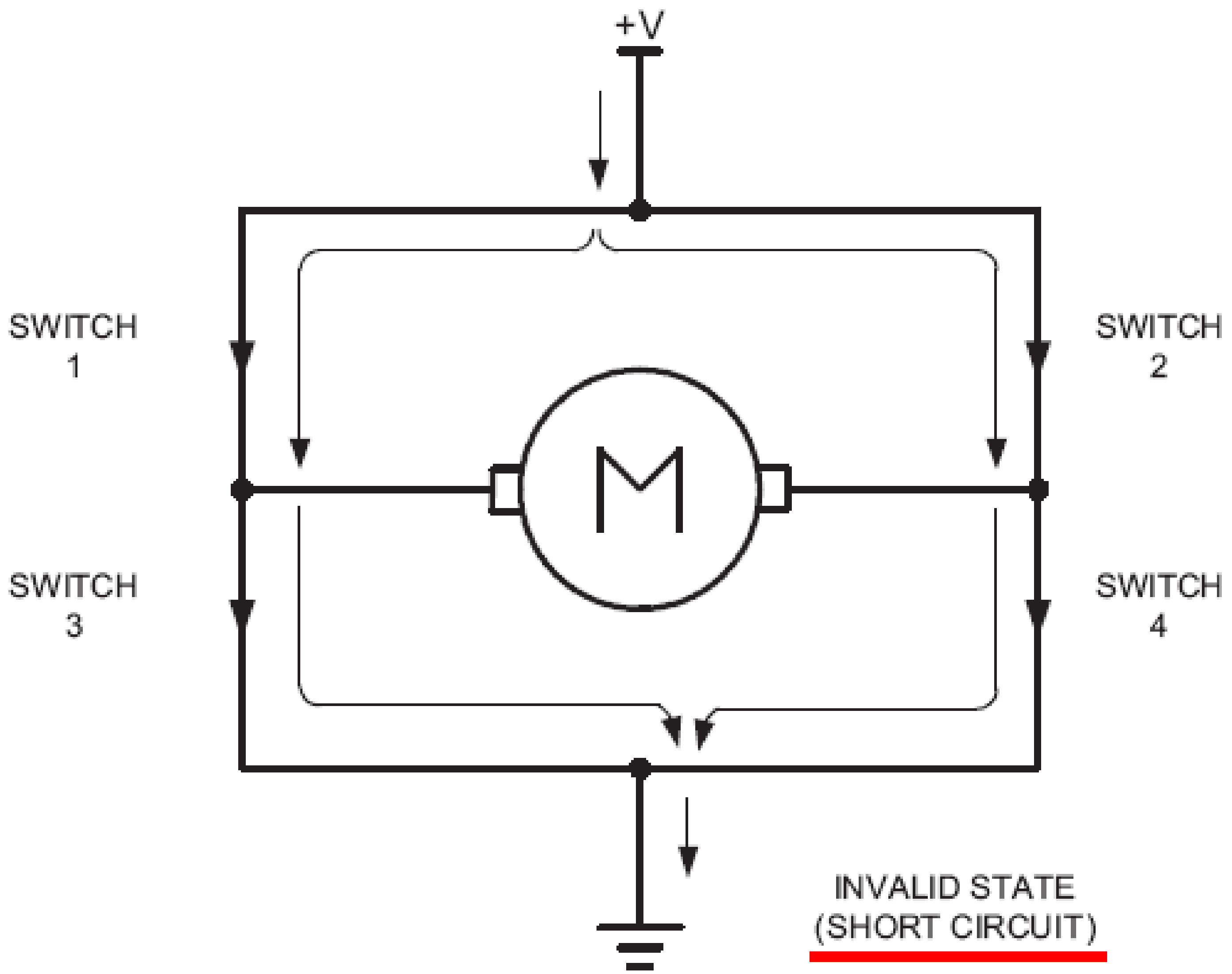
Motor DC control bidireccional



Motor DC control bidireccional



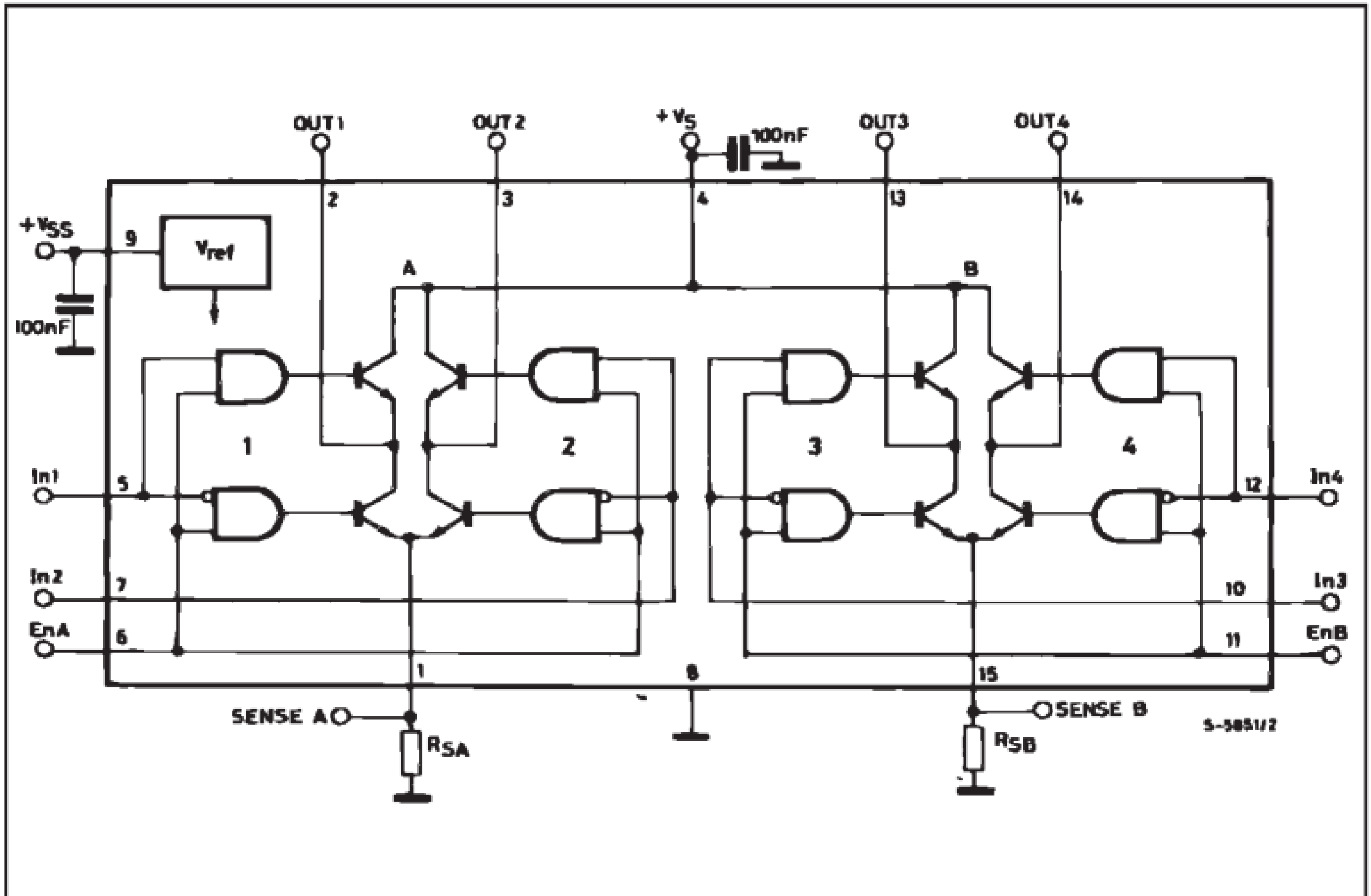
Motor DC control bidireccional (Invalid state)



PWM – Motor DC

- Control bidireccional con L298
- Posee 2 puentes H

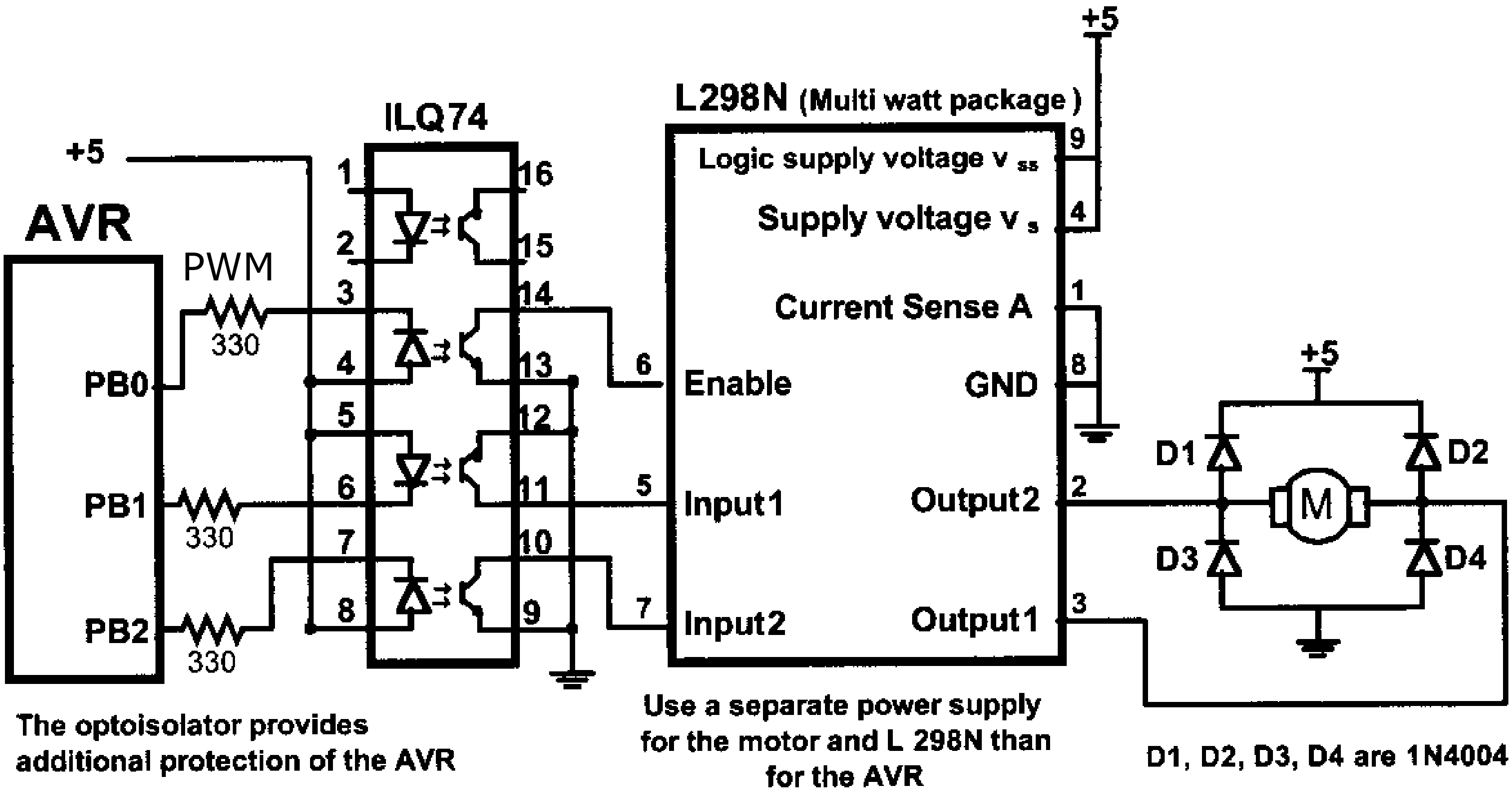
L298N Motor Controller Truth Table			
ENA	**IN1**	**IN2**	**Description**
0	N/A	N/A	Motor A is off
1	0	0	Motor A is stopped (brakes)
1	0	1	Motor A is on and turning backwards
1	1	0	Motor A is on and turning forwards
1	1	1	Motor A is stopped (brakes)



PWM – Motor DC

- Control bidireccional con L298

L298N Motor Controller Truth Table			
ENA	**IN1**	**IN2**	**Description**
0	N/A	N/A	Motor A is off
1	0	0	Motor A is stopped (brakes)
1	0	1	Motor A is on and turning backwards
1	1	0	Motor A is on and turning forwards
1	1	1	Motor A is stopped (brakes)

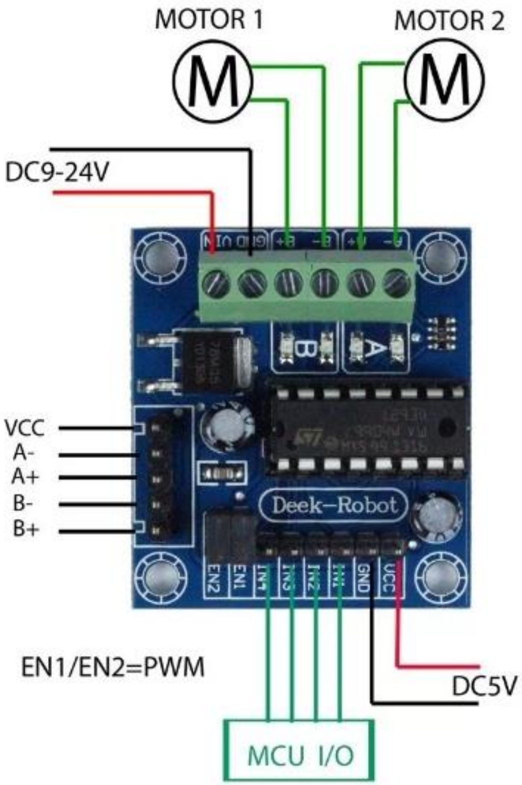


- PB1 y PB2 controlan la dirección de giro (Input 1 y 2) y PB0 controla la velocidad por PWM (Enable)

PWM – Motor DC

- Algunos ejemplos de módulos económicos

L298 (3A)



L293D (1A)

PWM Fase correcta– Timer 0

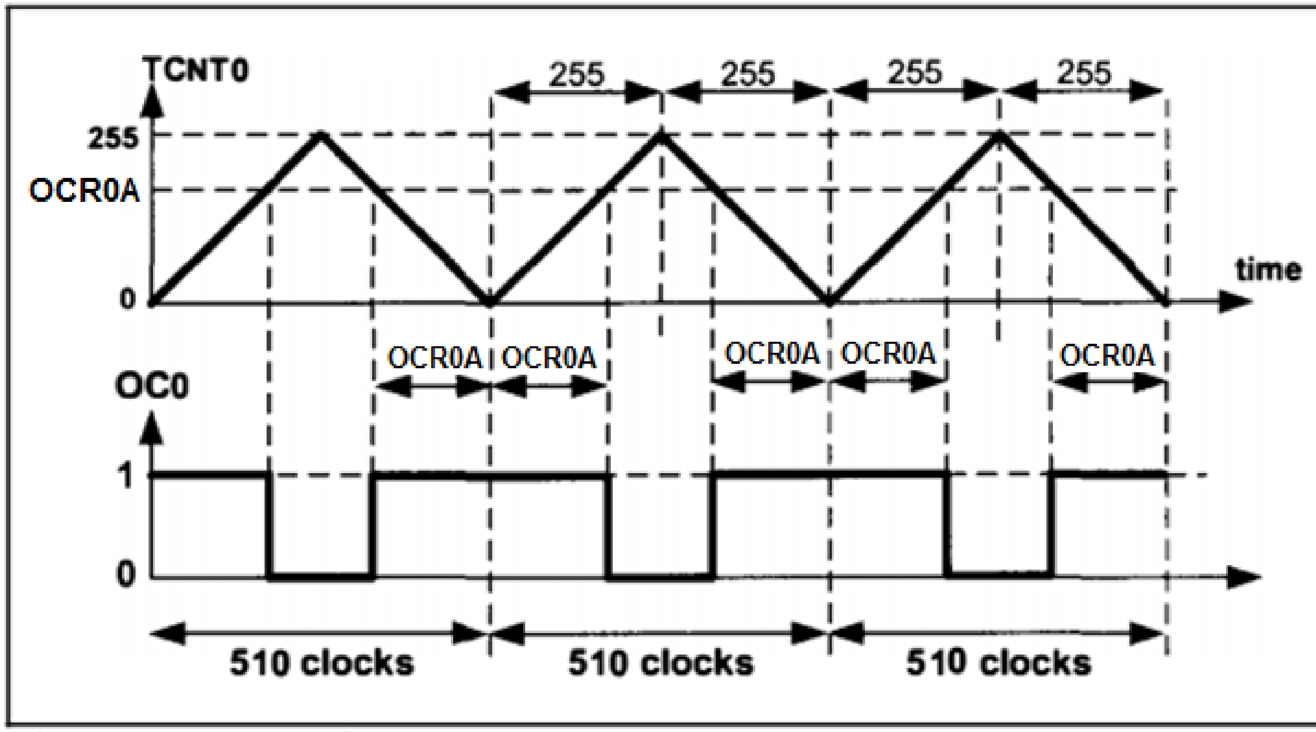
Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on[(1)(2)]
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

Notes:
1. MAX = 0xFF
2. BOTTOM = 0x00

PWM Fase correcta– Timer 0

- Modo Fase correcta

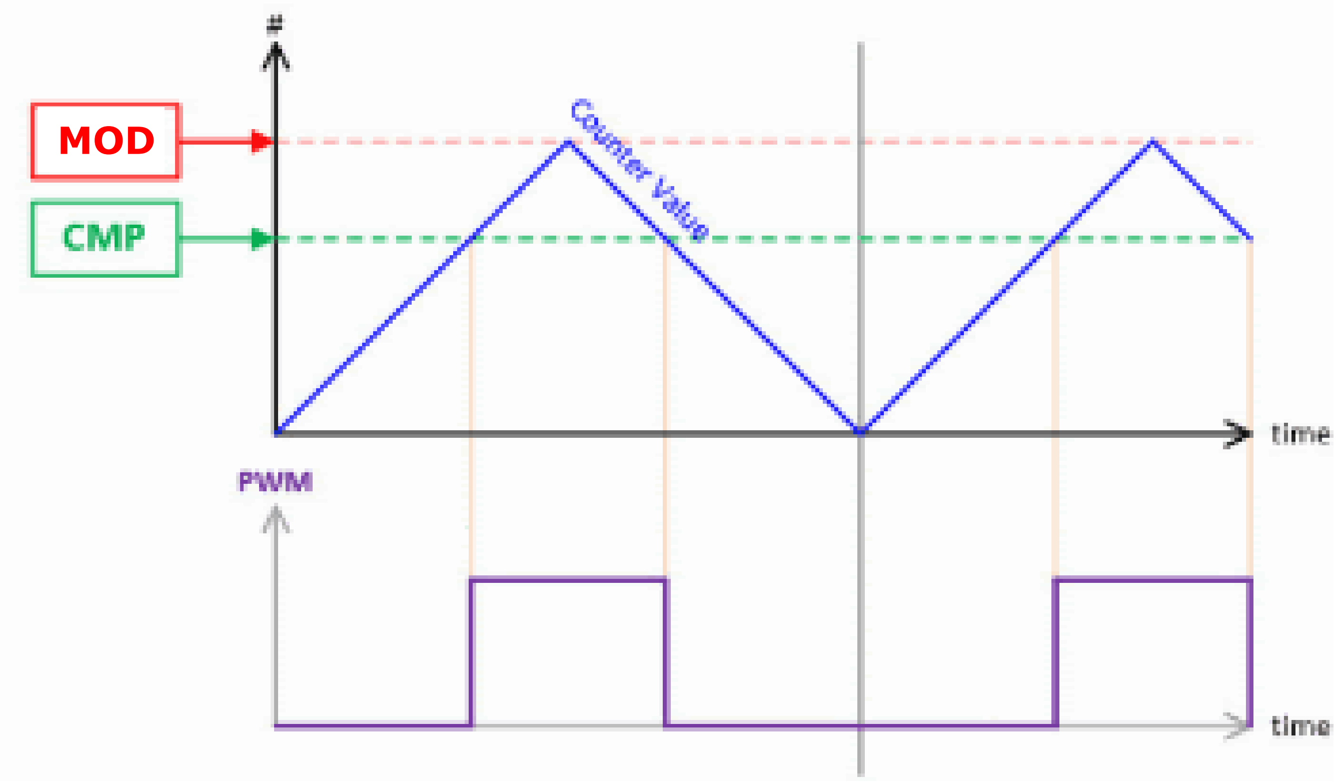


$$F_{\text{generated wave}} = \frac{F_{\text{timer clock}}}{510}$$

$$F_{\text{timer clock}} = \frac{F_{\text{oscillator}}}{N}$$

$$\Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{510 \times N}$$

PWM Fase correcta– Timer 0



PWM Fase correcta– Timer 1 (16 bits)

- Modos PWM

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	–	–	–
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

PWM Fase correcta – Timer 1 (16 bits)

- Modo Fase correcta

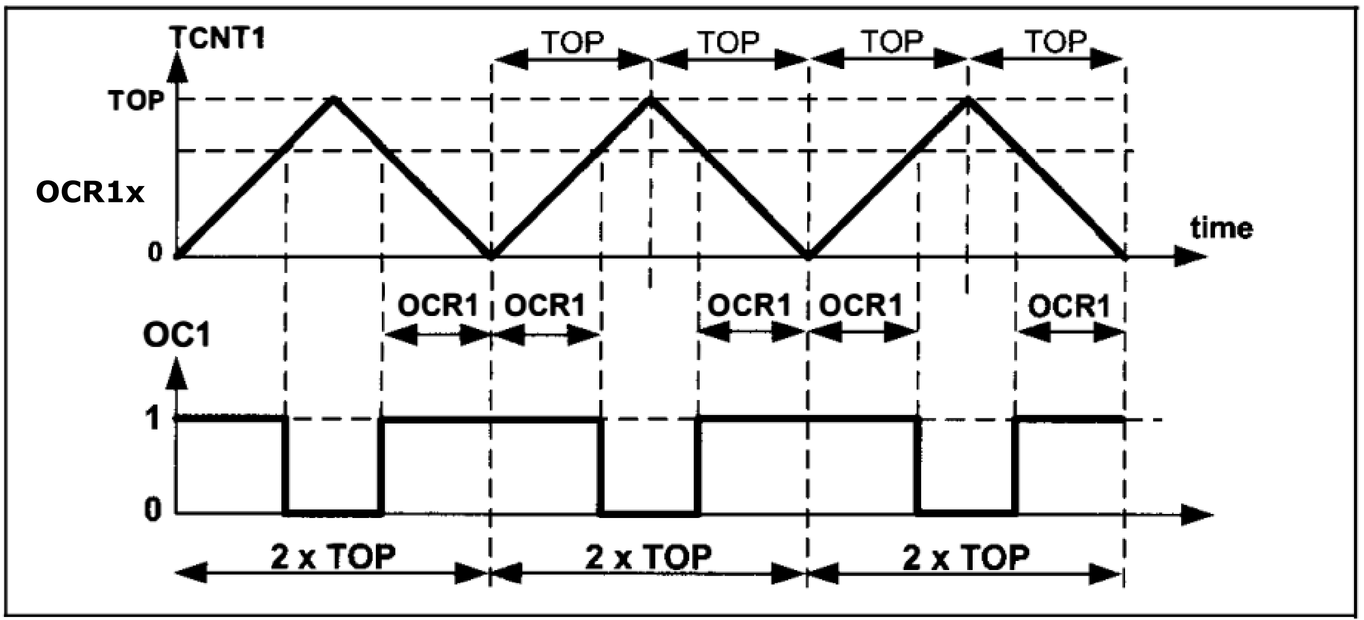


Figure 16-46. Timer/Counter 1 Phase Correct PWM Mode

$$F_{\text{generated wave}} = \frac{F_{\text{timer clock}}}{2 \times \text{Top}}$$

$$F_{\text{timer clock}} = \frac{F_{\text{oscillator}}}{N}$$

$$\Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{2 \times N \times \text{Top}}$$

PWM Fase correcta – Timer 1 (16 bits)

- Modo Fase correcta

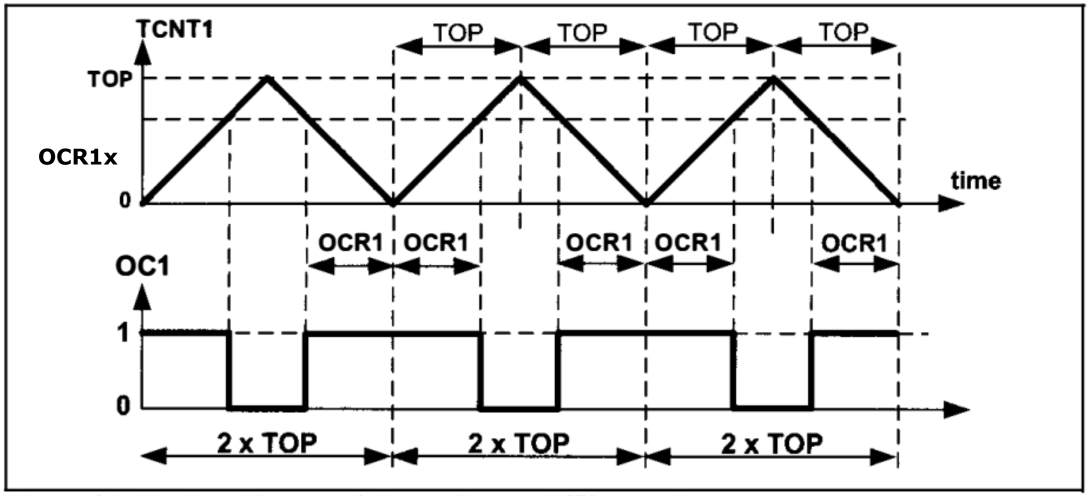


Figure 16-46. Timer/Counter 1 Phase Correct PWM Mode

- No Invertido:

$$\text{Duty Cycle} = \frac{2 \times \text{OCR1A}}{2 \times \text{Top}} \times 100 \Rightarrow \text{Duty Cycle} = \frac{\text{OCR1A}}{\text{Top}} \times 100$$

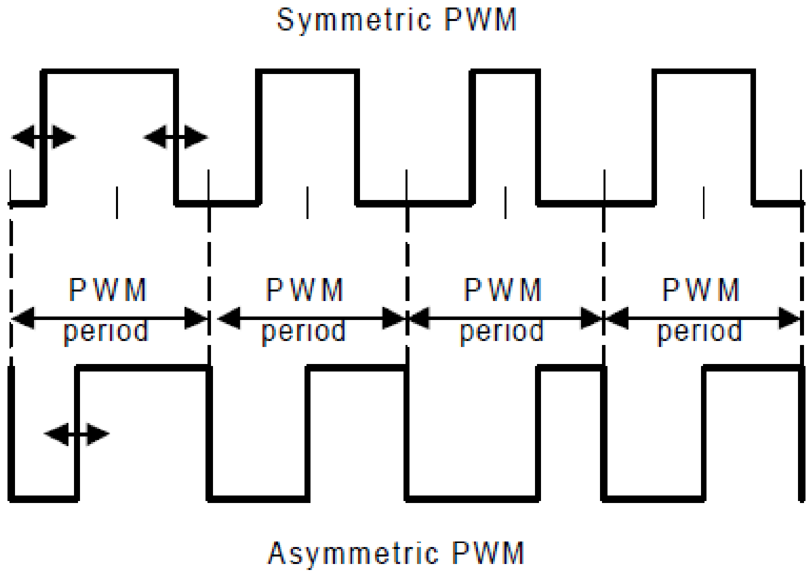
- Invertido:

$$\text{Duty Cycle} = \frac{2 \times \text{Top} - 2 \times \text{OCR1A}}{2 \times \text{Top}} \times 100 \Rightarrow \text{Duty Cycle} = \frac{\text{Top} - \text{OCR1A}}{\text{Top}} \times 100$$

Notar que puede obtenerse el rango 0 a 100% en un mismo modo

PWM Fase correcta – Timer 1 (16 bits)

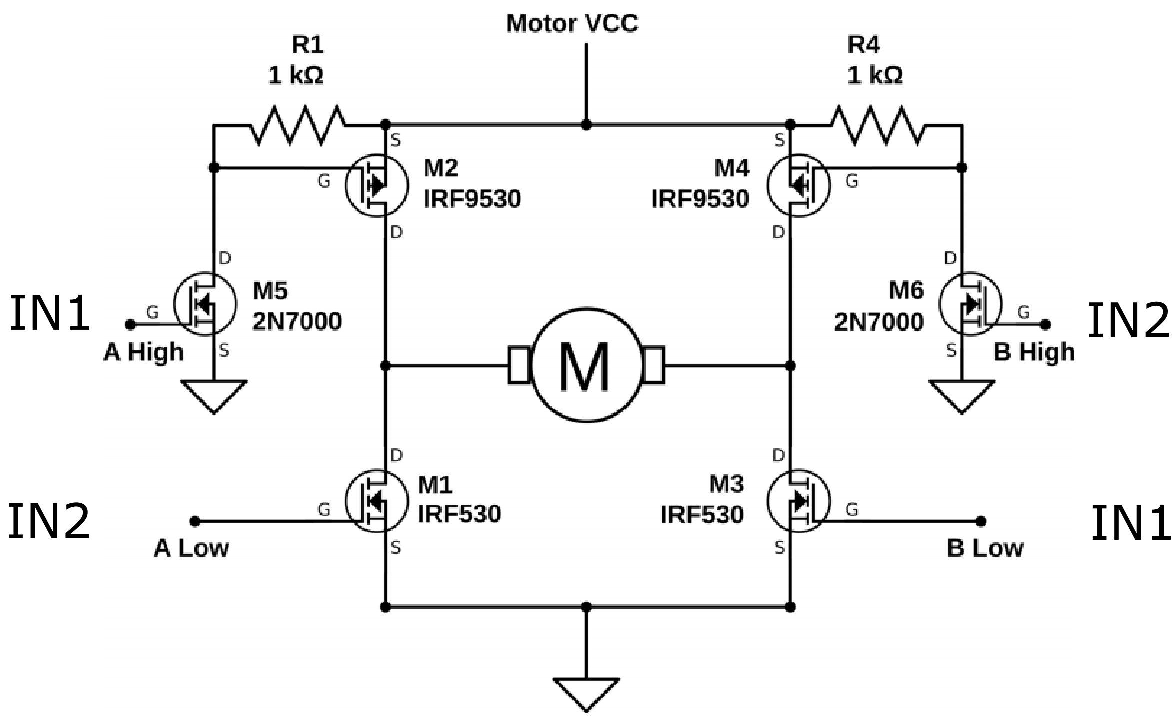
- Diferencias entre modos:



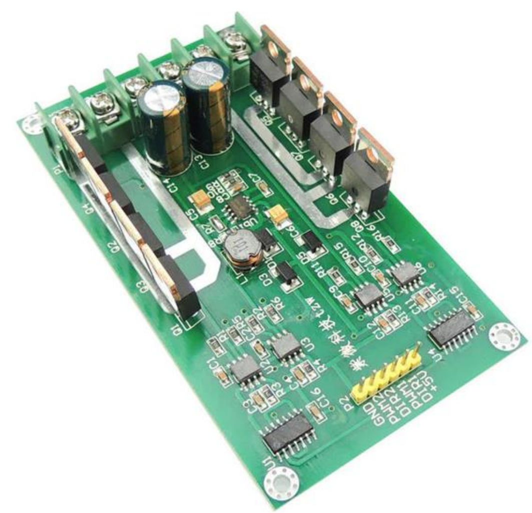
- Modo simétrico se utiliza preferentemente para el control de motores de múltiples fases como los motores de inducción o los BLDC (generalmente 3 o 6 fases).
- En fase correcta, el centro del pulso no varía, con lo cual puede utilizarse como instante para sensar la corriente.
- En motores de más de una fase el contenido armónico generado es menor.

PWM Fase correcta– Motor DC

- Ejemplo de aplicación: Control Bidireccional con Puente H
  - Cuando el uso de chips puente H no es posible por el alto consumo del motor, es necesario armar el circuito con transistores de potencia discretos.



Las señales IN1 e IN2
No se pueden activar
simultáneamente.
Se requiere de un zona de guarda
antes de la transición (Dead Zone)



PWM Fase correcta– Timer 1 (16 bits)

- En modo Fase correcta, la zona muerta de las señales PWM diferenciales puede controlarse fácilmente desplazando los valores de los registros:

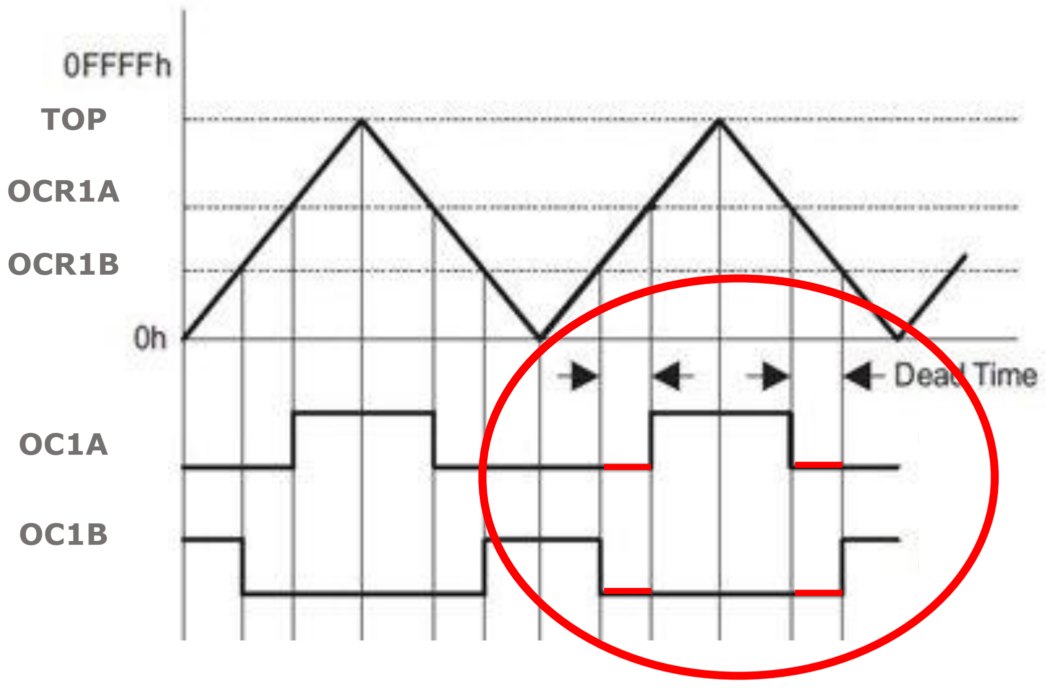
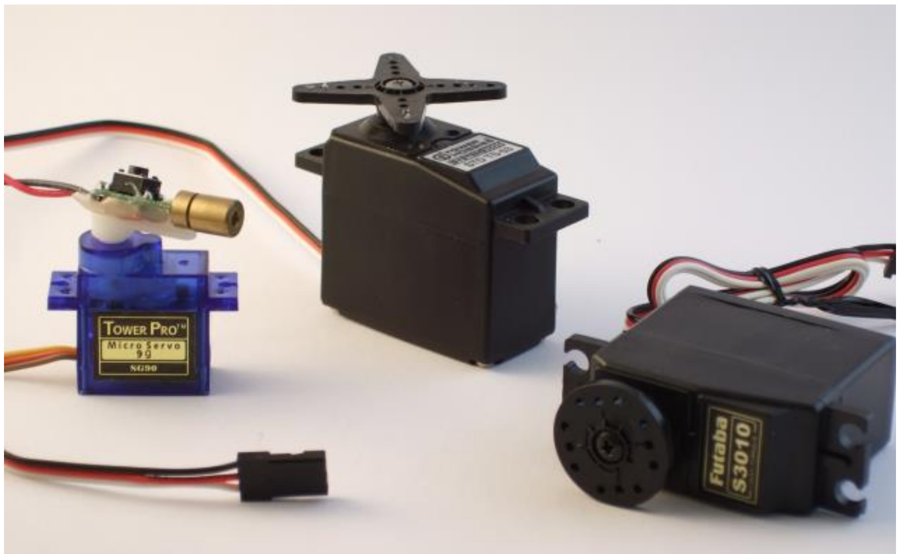
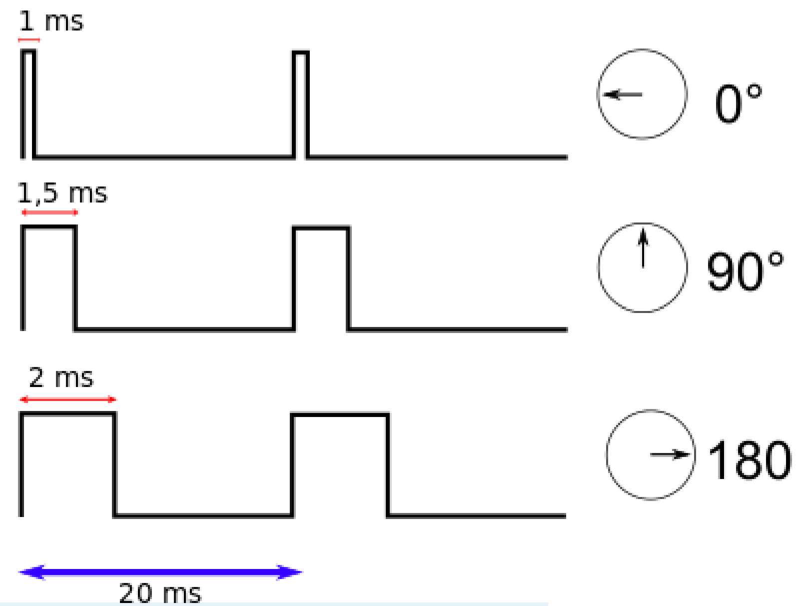


Figure 12-9. Output Unit in Up/Down Mode

OTRAS APLICACIONES

https://controlautomaticoeducacion.com/arduino/servomotor/



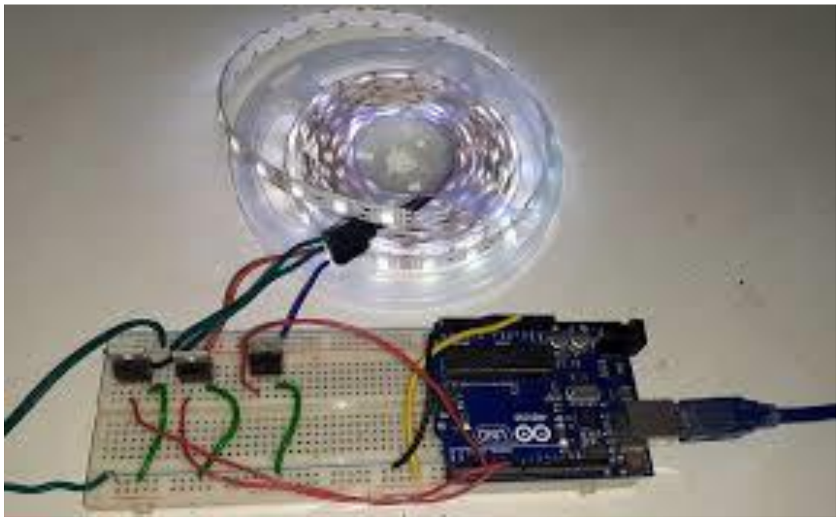


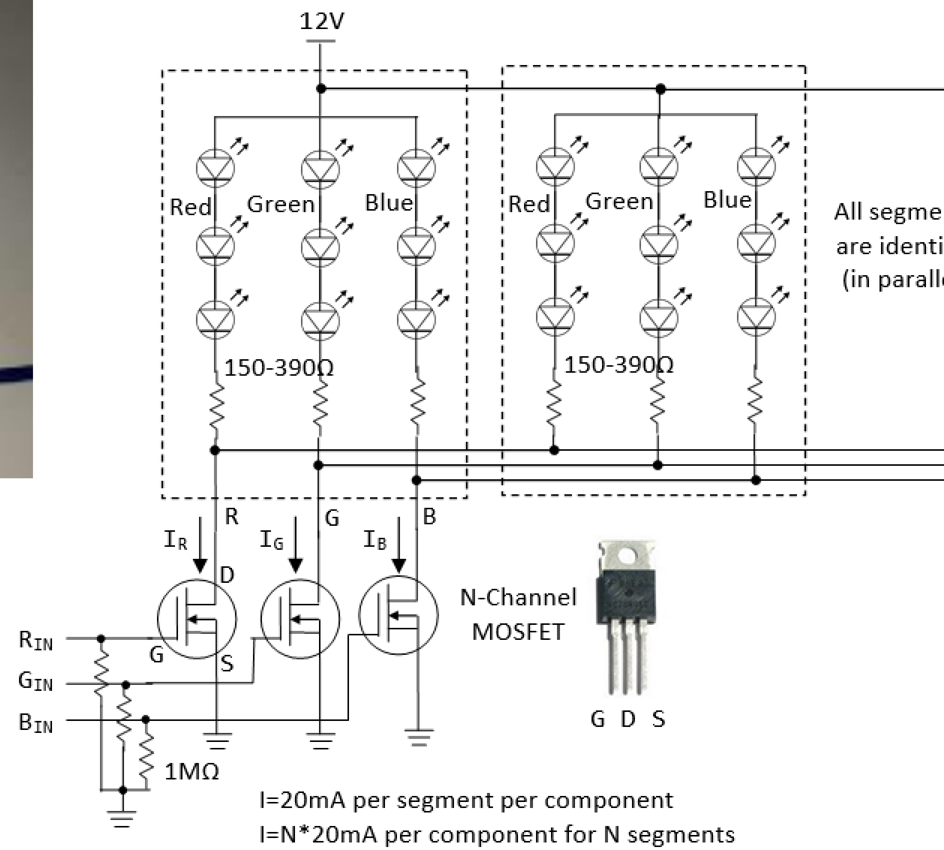

```c
static inline void initTimer1Servo(void) {
                        /* Set up Timer1 (16bit) to give a pulse every 20ms */
                                  /* Use Fast PWM mode, counter max in ICR1 */
  TCCR1A |= (1 << WGM11);
  TCCR1B |= (1 << WGM12) | (1 << WGM13);
  TCCR1B |= (1 << CS10);  /* /1 prescaling -- counting in microseconds */
  ICR1 = 20000;                                      /* TOP value = 20ms */
  TCCR1A |= (1 << COM1A1);              /* Direct output on PB1 / OC1A */
  DDRB |= (1 << PB1);                                /* set pin for output */
}
```

```c
  OCR1A = PULSE_MID;
  _delay_ms(1500);
```

OTRAS APLICACIONES

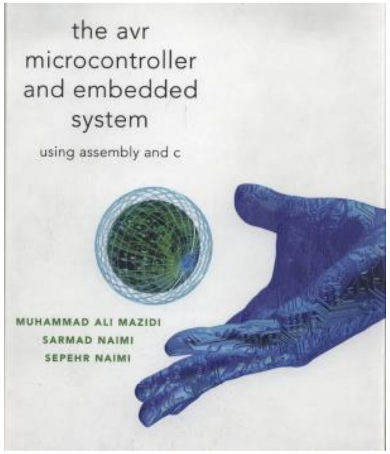
Circuito de control de Tiras de LEDs RGB

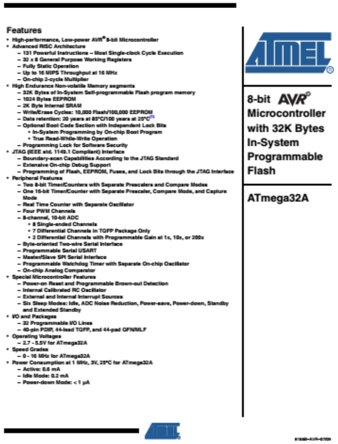




Bibliografía:

- *The AVR microcontroller & Embedded Systems*. Mazidi, Naimis, CH14,15 y16
- Libro Digital de Felipe Espinosa, CH4
- *DAC by using PWM: Using PWM to generate analog outpus, AN110*, June Choi, Hynix.
- *Using PWM Output as a Digital-to-Analog Converter on a TMS320F280x Digital Signal Controller, SPRAA88A*, David Alter, Texas Instruments, 2008.




Features

8-bit AVR Microcontroller with 32K Bytes In-System Programmable Flash

ATmega32A